The number of cycles in a graph: the second term, and pancyclic graphs

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Abstract

Let $\psi(r)$ be the largest number of cycles in a graph of cyclomatic number r. Entringer and Slater conjectured in 1981 that $\psi(r) \sim 2^{r-1}$. We prove that

$$\psi(r) \;\le\; 2^{r-1} + K\, r\, 2^{r/2}$$

for an absolute constant K and all $r \ge 1$. Graphs of Horak and Kim show that the order of the second term is correct. The constant K is not explicit, and the bound gives no numerical improvement for graphs of any practical size.

The paper has three ideas. First, count the cycles through a small subgraph W by their parts outside W; each such part is a forest. Second, about one half of all outside parts are forests, when the rank is large. This estimate gives $\psi(r) \sim 2^{r-1}$. It also gives $h(n) = \log_2 n + \log^* n + O(1)$ for the least number $n + h(n)$ of edges of a pancyclic graph on n vertices, which answers Erdős Problem #1016. Third, if the graph has large girth outside a short cycle W, then at most one half of the outside parts are forests, with no error term. This third idea is new. With two auxiliary theorems, on small cyclic cuts and on bounded girth, it gives the sharp bound.

The results on $\psi(r) \sim 2^{r-1}$ and on $h(n)$ are formalised in Lean 4. The sharp bound is not formalised. Parts of its proof are, and the paper says which. No referee has read this draft.

1 Introduction

1.1 The number of cycles

Graphs in this paper are finite and simple unless we say otherwise. Throughout, log is the binary logarithm, $\log^{(j)}$ is its j-fold iterate, and $\log^* x$ is the least $j \ge 0$ with $\log^{(j)} x \le 1$.

For a graph G, an *even subgraph* is a set of edges that meets every vertex an even number of times. With symmetric difference as addition, the even subgraphs form a vector space $\mathcal{Z}(G)$ over $\mathbb{F}_2$, the *cycle space*. Its dimension is

$$\beta(G) = |E(G)| - |V(G)| + \mathrm{comp}(G),$$

where $\mathrm{comp}(G)$ is the number of components. We call $\beta(G)$ the *rank* of G. Other names are cyclomatic number and cycle rank. A *cycle* is a connected 2-regular subgraph. We identify a subgraph that has no isolated vertices with its edge set. Then a cycle is a nonzero element of $\mathcal{Z}(G)$, so a graph of rank r has at most $2^r - 1$ cycles.

*Astra and Fable 5.2 are language models: GPT-6 Astra and Claude Fable 5.2. Section 1.5 says who did what, and what a machine has checked.

Let $\psi(r)$ be the largest number of cycles in a graph of rank r. Entringer and Slater [6] proved that $2^{r-1} < \psi(r) < 2^r$, and they conjectured that $\psi(r) \sim 2^{r-1}$. They also proved that the maximum is attained by a cubic graph for every $r \geq 3$. Aldred and Thomassen [2] proved that $\psi(r) \leq \frac{15}{16}2^r$, and that a planar graph of rank r has at most $2^{r-1} + o(2^{r-1})$ cycles. Alahmadi, Aldred, dela Cruz, Solé and Thomassen [1] proved that a graph with p vertices and $q > 2p + 3\log(3p)$ edges has at most 2^{q-p} cycles. For a connected graph, $2^{q-p} = 2^{r-1}$. From below, Horak and Kim [10] proved that

$$\begin{aligned} \psi(r) &\geq 2^{r-1} + (r - \tfrac{1}{2})2^{(r-1)/2} - \tfrac{3}{2}(r-1) \quad (r \text{ odd}), \\ \psi(r) &\geq 2^{r-1} + (r + \tfrac{5}{2})2^{(r-2)/2} - \tfrac{1}{2}(3r+2) \quad (r \text{ even}), \end{aligned} \tag{1.1}$$

for $r \geq 4$. Our main result shows that this is the right order of the second term.

Theorem 1.1 (Sharp bound)**.** *There is an absolute constant K such that every graph of rank $r \geq 1$ has at most*

$$2^{r-1} + K\, r\, 2^{r/2}$$

cycles. Consequently $\psi(r) = 2^{r-1} + \Theta(r\, 2^{r/2})$.

The theorem confirms the conjecture of Entringer and Slater with the correct second term. It is not a useful numerical bound. The constant K comes from a threshold R_0 on the rank, below which the proof uses only $\psi(r) < 2^r$. That threshold is not explicit and is very large. So $\frac{15}{16}2^r$ remains the best explicit bound for every rank of practical size. The theorem does not determine the best constant, and it does not determine $\psi(r)$.

The proof of Theorem 1.1 is long, and only parts of it are formalised. A weaker statement has a short proof from the forest estimate below, and this proof is formalised. It is also a step in the proof of Theorem 1.1.

Theorem 1.2 (Leading term)**.** *There is r_1 such that every graph of rank $r \geq r_1$ has at most*

$$\Big(\frac{1}{2} + \frac{21}{\log^{(3)} r}\Big)2^r$$

cycles. Consequently $\psi(r) \sim 2^{r-1}$.

1.2 The three ideas

First idea: outside parts. Fix a subgraph W of G. For an even subgraph ξ of G we call $\xi \setminus E(W)$ its *outside part*. If a cycle uses an edge of W, then its outside part is a proper subset of a cycle, so it is a forest. So the cycles through W are counted by the outside parts that are forests (Section 2). For an edge e of a cycle W, this gives a recursion: the cycles of G are the cycles of $G - e$ and the cycles through e, and the cycles through e are at most half of the even subgraphs whose outside part is a forest (Lemma 2.3).

Second idea: about one half of the outside parts are forests. This is an estimate of KNT [11].

Theorem 1.3 (Forest estimate, KNT [11])**.** *There is r_0 such that the following holds for all $r \geq r_0$. Let G be a connected simple graph of rank r. Let $W \subseteq G$ be a nonempty union of cycles with at most $\sqrt{r}$ edges and at most $(\log r)/100$ components. If ξ is a uniformly random element of $\mathcal{Z}(G)$, then*

$$\mathbb{P}(\xi \setminus E(W) \text{ is a forest}) \;\leq\; \frac{1}{2} + \frac{20}{\log^{(3)} r}. \tag{1.2}$$

The value $\frac{1}{2}$ has a simple source. Suppose that W meets the rest of G in two vertices u and v only. Then an outside part has even degree at every vertex, or it has odd degree exactly at u and v. There are equally many outside parts of each kind, and one of the first kind is not a forest unless it is empty (Example 2.2). Theorem 1.3 says that no other way to attach W leaves a larger proportion of forests, up to the error term.

Theorem 1.3 gives Theorem 1.2 (Section 3). It was found for another problem. A graph on n vertices is *pancyclic* if it contains a cycle of every length $3, 4, \dots, n$. Let $n + h(n)$ be the least number of edges of a pancyclic graph on n vertices.

Theorem 1.4 (Pancyclic graphs, KNT [11])**.** *There is an absolute constant A such that, for every $n \geq 3$,*

$$\log n + \log^* n - A \;\leq\; h(n) \;\leq\; \lfloor \log n \rfloor + \log^* n + 1.$$

Bondy [5] stated without proof that $\log(n-1) - 1 \leq h(n) \leq \log n + \log^* n + O(1)$. His lower bound is a count. A connected graph with $n + h$ edges has at most $2^{h+1} - 1$ cycles, and a pancyclic graph needs $n - 2$ of them. Griffin [8] wrote this count out and determined $h(n)$ for $n \leq 37$. Problem #1016 in Bloom's collection of Erdős problems [4] asks whether the upper bound is the truth, and whether even $h(n) - \log n \to \infty$. Theorem 1.4 answers both questions. The paper [11] gives more of the history.

Here is how the estimate enters the proof of Theorem 1.4. Let W have rank t and at least one edge, and sort the cycles of G by their outside parts. Two cycles with the same outside part differ only inside W, so the cycles with a given outside part have at most $|E(W)|$ different lengths. There are 2^{r-t} outside parts. Hence

$$\text{the cycles of } G \text{ have at most } |E(W)|\, 2^{r-t} \text{ different lengths.} \tag{1.3}$$

Theorem 1.3 improves this by a factor of almost two when r is large. Section 4 applies the improved bound at ranks that grow like a tower of exponentials. Each level halves the number of lengths that a given rank can supply, so each level costs one more edge. This gives the term $\log^* n$. Section 5 gives the construction for the upper bound.

Third idea: at most one half, at large girth. The error term in (1.2) cannot be removed in general. In the graphs of Horak and Kim a short cycle W has slightly more than half of its outside parts acyclic. The reason is local: these graphs have many cycles of length four, and many cuts of four edges. The third idea is that nothing else can go wrong. For a set P of vertices write

$$q(G, P) = \mathbb{P}\big(\xi \cap E(G - P) \text{ is a forest}\big).$$

If $P = V(W)$ for a subgraph W, then every edge of $G - P$ lies outside W, so $\mathbb{P}(\xi \backslash E(W) \text{ is a forest}) \leq q(G, P)$.

Theorem 1.5 (Exact half)**.** *There are absolute integers $D_* \geq 3$ and R with the following property. Let G be a connected cubic multigraph of rank $r \geq R$. Let P be a nonempty set of at most $3 \log r$ vertices such that $G[P]$ is connected and $G - P$ has no cycle of length less than D_*. Then*

$$q(G, P) \;\leq\; \frac{1}{2}.$$

The proof is by contradiction, and it does not use the size of the excess. Suppose that $q(G, P) > \frac{1}{2}$. The proof finds two disjoint parts of $G - P$ that are large, have small boundaries and have large girth. Given all edges outside the two parts, the two parts are independent, and each is a forest with

probability at most $\frac{3}{5}$. So $q(G,P) \le \frac{9}{25}$, a contradiction. The bound $\frac{3}{5}$ comes from Theorem 1.3, used once, with a fixed error. The two parts come from a chain of nested cycles, as follows. Most pairs of moderately long cycles have exactly two links in the sense of Section 6. Two links force one cycle to separate the other from P. Many such cycles around one point are nested, and the parts between them are the pieces that we need.

Theorem 1.5 applies to a shortest cycle W of a cubic graph of large girth. Then every edge of W lies on at most 2^{r-2} cycles, and the recursion of the first idea gives $\psi(r) \le \psi(r-1) + 2^{r-2}$ for such graphs. Two further theorems treat the graphs that Theorem 1.5 excludes. One is for cubic graphs with a cyclic cut of at most four edges (Theorem 7.1). The other is for cubic graphs of bounded girth without such a cut (Theorem 7.2). Section 7 states them and derives Theorem 1.1. Their proofs are long case analyses of a different kind. They are in a supplement to this paper (Section 12).

1.3 What the third idea does for the two older results

We asked whether Theorem 1.1 or its proof improves Theorem 1.2 or Theorem 1.4. The answers are different.

The number of cycles. Theorem 1.1 replaces the statement of Theorem 1.2 by a sharp one. It does not replace its proof. The two auxiliary theorems use the bound $(\frac{1}{2} + o(1))2^r$ as an input, so Theorem 1.2 is the first step of the proof of Theorem 1.1.

Pancyclic graphs. Theorem 1.1 does not change Theorem 1.4. The term $\log n$ in $h(n)$ comes from the number of cycles, and Theorem 1.1 sharpens only the error in that count: a pancyclic graph of rank r on n vertices has $n - 2 \le 2^{r-1} + Kr2^{r/2}$. The term $\log^* n$ comes from the lengths of the cycles. It needs the forest estimate for a subgraph W that is a union of many short cycles, in a graph that contains short cycles everywhere. Theorem 1.5 does not apply to such graphs. A pancyclic graph contains cycles of length three and four, and the proportion of forests can exceed $\frac{1}{2}$ when such cycles are present (Example 2.2 and Section 13).

One tool from the proof does apply to Theorem 1.3. The parallel-link lemma (Lemma 10.1) bounds one term of the link count without the error term of (1.2). With it, the error term $20/\log^{(3)} r$ in Theorem 1.3 improves to $r^{-\kappa}$ for every fixed $\kappa < \frac{1}{3}$, for unions of cycles with at most $r/(2\log^2 r)$ edges and at most $\gamma(\kappa)\log r$ components. A further refinement gives $O((\log r)^{11/2}/\sqrt{r})$ (Remark 11.2). This would shorten the error analysis in Section 4. It would not change the statement of Theorem 1.4, because the lower bound there already loses only an additive constant.

1.4 Two elementary theorems

Two further results concern graphs of small rank. Their proofs are elementary and do not use Theorem 1.3.

Theorem 1.6. (a) *Let N be a connected graph and let $u \ne v$ be two of its vertices. Then N has at most $2^{\beta(N)}$ paths from u to v. If it has exactly $2^{\beta(N)}$, then each of these paths has at least $\beta(N)$ edges, and deleting the edges of any one of them leaves a forest.*

(b) *Let G be a connected graph and let $W \subseteq G$ be a subgraph that contains a cycle. Suppose that every outside part is the outside part of some cycle of G. Then $G - E(W)$ has no two edge-disjoint cycles, and $\beta(G) \le \beta(W) + |V(W)| + 3$.*

Part (a) is about graphs that attain the largest possible number of paths between two vertices. It explains the shape of the construction in Section 5. A gadget that turns k units of rank into 2^k path lengths has only long paths, so the short cycles must come from a second, much smaller

gadget, and so on. This is the source of the term $\log^* n$ in the construction. The bound of part (a) is also a tool in the supplement: the proofs there count paths between two vertices, and they need a stable form of part (a) (Theorem 12.1).

Part (b) is about the count of cycles by their outside parts. Call an outside part *wasted* if no cycle has it as its outside part. Part (b) says that a graph can avoid all waste only while its rank exceeds the rank of W by at most $|V(W)| + 3$. Section 14 proves both parts.

1.5 Authorship and verification

KNT is a person. Astra and Fable 5.2 are language models. Theorems 1.3 and 1.4, their proofs, and the Lean formalisation of these proofs come from [11]. KNT obtained them in September 2026 with Astra, and has described that process in public [12]. In November 2026 KNT asked Fable 5.2 to look for a simpler proof of Theorem 1.3. That search found Theorems 1.2 and 1.6, and Fable 5.2 wrote their Lean proofs. KNT then asked both models to study the second term of $\psi(r)$, in one shared set of files and in parallel. Theorem 1.1 is the result of that work. KNT directed the work at each step.

The parts of the proof of Theorem 1.1 have these origins. The edge recursion, the chain of nested cycles, the parallel-link lemma and the first form of the localisation lemma are due to Fable 5.2. The stable form of Theorem 1.6(a), the two auxiliary theorems, the bound $\frac{3}{5}$ for the pieces, the packing of short cycles without an excess, the use of two girth scales, and the assembly of Theorem 1.5 are due to Astra. Each model checked the proofs of the other in the parts that this paper prints. Fable 5.2 wrote this paper. For Sections 7 to 11 it followed Astra's written proof [18] and checked every step. It did not check the supplement line by line (Section 12).

Statement	Status	Lean declaration (namespace `Erdos1016`)
Theorem 1.1	not formalised	parts only; see Table 3
Theorem 1.5	not formalised	parts only; see Table 3
Theorem 1.2, upper bound	formalised	`cycleCountTheorem`, `cycleCountTheorem_general`
Theorem 1.2, $\psi(r) \sim 2^{r-1}$	formalised	`cycleCountAsymptotic`
Theorem 1.3	formalised	`ShortProof.fewComponentForestEstimate`
Theorem 1.4	formalised	`mainTheorem`; the explicit upper bound is `Problem1016.upper_bound_integer`
Theorem 1.6(a)	formalised	`twoTerminalTheorem`
Theorem 1.6(b)	formalised	`noWasteTheorem`

Table 1: The theorems and their formal counterparts in the repository [13]. Appendix A names the commit.

Theorems 1.2, 1.3, 1.4 and 1.6 are formalised in Lean 4 [15] with Mathlib [17]. Table 1 lists the declarations. The Lean kernel accepts their proofs with the axioms `propext`, `Classical.choice` and `Quot.sound` only. This certifies the formal statements. It does not certify the prose of this paper. Appendix A prints the formal statements of Theorems 1.2 and 1.6, so that the reader can compare them with the text. It also gives the commands that repeat the check and names the Lean declaration for each step.

Theorems 1.1 and 1.5 are not formalised. Their proofs have been read by the two models only. Some steps of the proof of Theorem 1.5 are formalised: the parallel-link lemma, the link bound, the moment inequality, and the order of nested separators. Table 3 in Appendix A lists them. The reader should treat Theorem 1.1 as a claim with a written proof, and not as a verified result.

The examples and the remarks in this paper are not formalised. Appendix B describes tests of the examples, and of our wording of the proofs, on small graphs. Appendix C says which sources we read and how we searched the literature. No referee has read this draft.

2 Outside parts

Let W be a subgraph of a graph G. Put $r = \beta(G)$, $t = \beta(W)$ and $m = |E(W)|$. The map $\xi \mapsto \xi \setminus E(W)$ is linear on $\mathcal{Z}(G)$, and its kernel is $\mathcal{Z}(W)$. So there are 2^{r-t} outside parts, and each of them is the outside part of exactly 2^t even subgraphs. If ξ is uniform on $\mathcal{Z}(G)$, then its outside part is uniform on the set of outside parts. Let

$$p(G, W) = \mathbb{P}(\xi \setminus E(W) \text{ is a forest}).$$

This is the proportion of outside parts that are forests. It is also the proportion of even subgraphs whose outside part is a forest:

$$\big|\{\xi \in \mathcal{Z}(G) : \ \xi \setminus E(W) \text{ is a forest}\}\big| \ = \ p(G, W)\, 2^r. \tag{2.1}$$

Lemma 2.1 (Counting lemma)**.** *Let $p = p(G, W)$.*

(a) *If a cycle C of G contains an edge of W, then $C \setminus E(W)$ is a forest. Hence at most $p\, 2^r$ cycles of G contain an edge of W.*

(b) *The cycles of G have at most $2^{r-t}(pm + 1 - p)$ different lengths.*

Proof. (a) A proper subset of the edge set of a cycle contains no cycle. The second statement follows from (2.1), because distinct cycles are distinct even subgraphs.

(b) Let x be an outside part. Suppose first that x is not a forest, and let C be a cycle with outside part x. Then x contains a cycle and $x \subseteq C$, which forces $C = x$. So x accounts for at most one cycle, and hence for at most one length. Now suppose that x is a forest, and let C be a cycle with outside part x. Then C contains an edge of W, because C is not a forest. The length of C is $|x| + |C \cap E(W)|$, and $1 \le |C \cap E(W)| \le m$. So x accounts for at most m lengths. There are $p\, 2^{r-t}$ outside parts that are forests and $(1 - p)\, 2^{r-t}$ that are not. □

If W has at least one edge, then part (b) with $p \le 1$ gives the bound (1.3) of the introduction. The lemma improves on it when p is less than one.

Example 2.2 (Two points of attachment)**.** Suppose that $G = W \cup N$, where W and N are connected, have no common edge, and have exactly two common vertices u and v. A $\{u, v\}$-*join* of N is a set of edges of N in which u and v have odd degree and all other vertices have even degree. The outside parts are the $2^{\beta(N)}$ even subgraphs of N and the $2^{\beta(N)}$ joins of N. A nonempty even subgraph is not a forest. Hence

$$p(G, W) \ \le \ \frac{1 + 2^{\beta(N)}}{2^{\beta(N)+1}} \ = \ \frac{1}{2} + 2^{-\beta(N)-1},$$

with equality if and only if every join of N is a forest. So when W is attached at two vertices, the error term in (1.2) can be replaced by $2^{-\beta(N)-1}$, which is exponentially small.

For a concrete case, let N be a chain of k four-cycles, each joined to the next at opposite vertices, and let u and v be the two ends of the chain. Every join of N is a path from u to v, so equality

holds. If W is a triangle and k is large, this graph satisfies the hypotheses of Theorem 1.3. So the constant $\frac{1}{2}$ in (1.2) cannot be lowered.

Figure 1 shows the case $k = 2$. There $\beta(G) = 4$ and $\beta(N) = 2$. The eight outside parts are the four even subgraphs of N and the four paths from u to v in N. Five of them are forests, so $p(G, W) = 5/8$. Lemma 2.1(a) allows $\frac{5}{8} \cdot 16 = 10$ cycles through W, and there are nine.

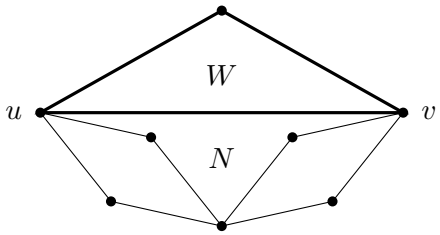


Figure 1: The graph of Example 2.2 for $k = 2$: a triangle W (thick) and a chain N of two four-cycles.

One edge of a cycle. The next lemma is the form of the count that Theorem 1.1 needs. It holds for multigraphs, with the same proof. For an edge e let $c(G)$ be the number of cycles of G and $c_e(G)$ the number of cycles of G through e. Recall that $q(G, P) = \mathbb{P}(\xi \cap E(G - P)$ is a forest) for a set P of vertices.

Lemma 2.3 (Edge recursion)**.** *Let G be a connected graph of rank r, let W be a cycle of G, and let e be an edge of W. Then $G - e$ is connected and has rank $r - 1$, and*

$$c(G) = c(G - e) + c_e(G), \qquad c_e(G) \;\le\; q(G, V(W))\, 2^{r-1}.$$

Proof. The edge e lies on a cycle, so $G - e$ is connected, and it has one edge less. A cycle of G either avoids e or contains e. This is the first formula.

Let C be a cycle through e. Then C and $C + W$ are even subgraphs of G. The cycle C has a vertex in $V(W)$, so $C \cap E(G - V(W))$ is a proper part of a cycle, which is a forest. Every edge of W has both ends in $V(W)$, so $C + W$ has the same edges in $G - V(W)$ as C. The even subgraph C contains e, and $C + W$ does not. So if C' is another cycle through e, the four sets C, $C + W$, C', $C' + W$ are distinct. Hence there are at least $2c_e(G)$ even subgraphs whose part in $G - V(W)$ is a forest, and their number is $q(G, V(W))\, 2^r$. □

In particular, if $q(G, V(W)) \le \frac{1}{2} + \varepsilon$, then $c(G) \le c(G - e) + (\frac{1}{4} + \frac{\varepsilon}{2})2^r$. For a graph with the largest number of cycles this reads $\psi(r) \le \psi(r - 1) + (\frac{1}{4} + \frac{\varepsilon}{2})2^r$. If ε is zero, the difference $\psi(r) - 2^{r-1}$ does not grow at all in this step.

3 The number of cycles

This section proves Theorem 1.2. The main step is Lemma 3.2. Appendix A names the Lean declaration for each step.

3.1 The lower bound

Entringer and Slater proved that $\psi(r) > 2^{r-1}$ for $r \geq 3$. Their examples are cubic: a cycle of even length with all its long diagonals. These graphs give $\psi(r) \geq 2^{r-1} + r^2 - 3r + 3$ [6, Corollary 4].

A simpler example suffices here. Let $j \geq 2$. Take vertices $u_1, \ldots, u_j$ and, for each i, two further vertices x_i and y_i that are joined to u_i and to u_{i+1}, with indices modulo j. This is a ring of j four-cycles: a chain like N in Figure 1, closed up. It has $3j$ vertices and $4j$ edges, so its rank is $j + 1$. Its cycles are the j four-cycles and the 2^j cycles that pass through all the vertices u_i. So $\psi(r) \geq 2^{r-1} + r - 1$ for $r \geq 3$. The formal proof uses a closed chain of shortcuts from the construction of Section 5 instead, which has rank $j + 1$ and at least 2^j cycles.

3.2 The upper bound

In this subsection G is a connected simple graph of rank r, and k is an integer with $1 \leq k \leq r$.

Lemma 3.1 (Independent edges)**.** *There is a set D of k edges of G such that exactly 2^{r-k} even subgraphs of G contain no edge of D. Every edge of D lies on a cycle of G.*

Proof. Fix a spanning tree of G. Exactly r edges of G lie outside the tree. Let D consist of k of them. The graph $G - D$ contains the tree, so it is connected, and its rank is $r - k$. The even subgraphs of G that contain no edge of D are the even subgraphs of $G - D$. So there are 2^{r-k} of them. An edge of D lies on a cycle, because the tree contains a path between its ends. □

Chords and triangles. Let $e = uv$ be an edge of D. It lies on a cycle, so v has a neighbour $w_e \neq u$. Fix such a neighbour for each edge of D. If u and w_e are not adjacent in G, add the edge uw_e and call it a *chord* (Figure 2). Let G^+ be the graph that we get when we add all chords, each of them once, and let $a \leq k$ be the number of chords. The graph G^+ is simple and connected, and it has the same vertices as G. So $\beta(G^+) = r + a$.

For $e \in D$ let T_e be the triangle of G^+ with the edges uv, vw_e and uw_e, and let W be the union of these triangles. Then W is a nonempty union of cycles of G^+, with at most $3k$ edges and at most k components. It contains every edge of D and every chord. Every chord f lies on a triangle T_f of W whose other two edges are edges of G. If there are several such triangles, fix one of them.

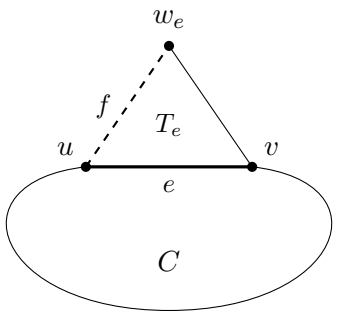


Figure 2: An edge $e = uv$ of D, a second neighbour w_e of v, and the chord $f = uw_e$ (dashed). The chord is added if u and w_e are not adjacent in G. A cycle C through e and the even subgraph $C + T_e$ have the same edges outside the triangle T_e.

Lemma 3.2 (Cycles through D)**.** *At most $p(G^+, W)\, 2^r$ cycles of G contain an edge of D.*

Proof. Let F be the set of chords, so $|F| = a$. For a cycle C of G that contains an edge of D, and for a set $S \subseteq F$, put

$$C_S = C + \sum_{f \in S} T_f.$$

This is an even subgraph of G^+. We show two things.

The outside part of C_S is a forest. All triangles lie in W, so $C_S \setminus E(W) = C \setminus E(W)$. The cycle C contains an edge of D, and $D \subseteq E(W)$. So $C \setminus E(W)$ is a forest by Lemma 2.1(a).

The even subgraph C_S determines the pair (C, S). The cycle C contains no chord. The triangle T_f contains the chord f and no other chord. So the set of chords in C_S is S. Thus C_S determines S, and then it determines $C = C_S + \sum_{f \in S} T_f$.

Let q be the number of cycles of G that contain an edge of D. The sets C_S are $q\,2^a$ distinct even subgraphs of G^+ whose outside part is a forest. By (2.1), applied to G^+ and W, there are exactly $p(G^+, W)\,2^{r+a}$ such even subgraphs. Hence $q \le p(G^+, W)\,2^r$. □

Proof of Theorem 1.2. Let G be a graph of rank r, where r is large. If G is not connected, add a new vertex and join it to one vertex of each component. The new edges are bridges, so the rank and the set of cycles do not change. So we may assume that G is connected.

Let k be the least integer with $2^k > \log r$. Then $k \le \log\log r + 1$, and $1 \le k \le r$. Take D as in Lemma 3.1, and take G^+ and W as above. A cycle of G avoids D or contains an edge of D. At most $2^{r-k} < 2^r/\log r$ cycles avoid D, by Lemma 3.1. At most $p(G^+, W)\,2^r$ cycles contain an edge of D, by Lemma 3.2.

It remains to bound $p(G^+, W)$. The graph G^+ is connected and simple, and its rank $r^+ = r + a$ is at least r. The subgraph W is a nonempty union of cycles. It has at most $3k \le \sqrt{r} \le \sqrt{r^+}$ edges and at most $k \le (\log r)/100 \le (\log r^+)/100$ components, because r is large. So Theorem 1.3 applies to G^+ and W, and

$$p(G^+, W) \;\le\; \frac{1}{2} + \frac{20}{\log^{(3)} r^+} \;\le\; \frac{1}{2} + \frac{20}{\log^{(3)} r}.$$

Altogether, the number of cycles of G is at most

$$\Big(\frac{1}{2} + \frac{20}{\log^{(3)} r} + \frac{1}{\log r}\Big)2^r \;\le\; \Big(\frac{1}{2} + \frac{21}{\log^{(3)} r}\Big)2^r,$$

because $\log^{(3)} r \le \log r$. Together with the lower bound, this gives $\psi(r) \sim 2^{r-1}$. □

Remark 3.3. The choice of k is not delicate. The proof needs $2^{-k} \le 1/\log^{(3)} r$ for the cycles that avoid D, and $100k \le \log r$ for the components of W.

The chords are the point of the proof. Without them one would take for W a union of cycles of G through the edges of D. Such cycles can be long, and Theorem 1.3 needs W to be small. To shorten them one would have to pass to a cubic multigraph and search for short cycles there. The chords create the short cycles directly, and Lemma 3.2 shows that they cost nothing.

4 The lower bound for pancyclic graphs

This section derives the lower bound of Theorem 1.4 from Theorem 1.3. The plan is as follows. A number $\Phi(R) \le 1$ measures how well a graph of rank at least R can fill an interval of cycle lengths. In a graph with many cycle lengths we find a small subgraph W that already contains the short

cycles. The forest estimate for W then shows that Φ drops by a factor of almost two when the rank grows from R to 2^{256R}.

For an integer $R \geq 1$ let

$$\Phi(R) = \sup \frac{L-2}{2^{\beta(G)}},$$

where the supremum is over all simple graphs G with $\beta(G) \geq R$ and all integers $L \geq 2$ such that G contains a cycle of each length $3, \dots, L$. A graph of rank β has at most $2^\beta - 1$ cycles, so $\Phi(R) \leq 1$. The function Φ is non-increasing, because a larger R allows fewer graphs.

Lemma 4.1 (Witness)**.** *Let $R \geq 1$ and $\varphi = \Phi(R)$, and let G contain a cycle of each length $3, \dots, L$, where $L \geq 2^{2R}$. Then G has a subgraph W that is a union of at most $2R$ cycles, with $\beta(W) \geq 2R$ and $|E(W)| \leq \varphi\, 2^{2R} + 2^R + 5R$.*

Proof. Put $W_0 = \emptyset$ and $W_{i+1} = W_i \cup C_i$, where C_i is a cycle of G whose length ℓ_i is the least integer ≥ 3 that is not the length of a cycle of W_i. Stop as soon as the rank is at least $2R$. Consider a step, and let $s = \beta(W_i) < 2R$. The graph W_i contains cycles of the $\ell_i - 3$ lengths $3, \dots, \ell_i - 1$. Therefore $\ell_i - 3 \leq 2^s - 1$. This gives $\ell_i \leq 2^{2R-1} + 2 \leq L$, so C_i exists. If $s \geq R$, then also $\ell_i - 3 \leq \varphi\, 2^s$, by the definition of Φ. The cycle C_i is not a subgraph of W_i, so the rank increases at every step. So there are at most $2R$ steps, their ranks s are distinct, and

$$|E(W)| \;\leq\; \sum_i \ell_i \;\leq\; \sum_{s=0}^{R-1} (2^s + 2) + \sum_{s=R}^{2R-1} (\varphi\, 2^s + 3) \;\leq\; \varphi\, 2^{2R} + 2^R + 5R. \qquad \square$$

Proposition 4.2 (Near-halving)**.** *For every sufficiently large integer R,*

$$\Phi\big(2^{256R}\big) \;\leq\; \Big(\frac{1}{2} + \frac{20}{\log^{(2)} R}\Big)\Phi(R) + 2^{2-R}.$$

Proof. Let G be a simple graph of rank $r \geq 2^{256R}$ with a cycle of each length $3, \dots, L$. We bound $(L-2)\, 2^{-r}$. Identifying a vertex of one component with a vertex of another changes neither the rank nor the set of cycles, so we may assume that G is connected. If $L < 2^{2R}$, then $(L-2)\, 2^{-r} < 2^{2-R}$, and we are done.

Otherwise let W be as in Lemma 4.1, with m edges and rank $t \geq 2R$, and let $\varphi = \Phi(R)$. Then $m \leq 2^{2R+1} \leq \sqrt{r}$ and $\mathrm{comp}(W) \leq 2R \leq (\log r)/100$. This is the only place in this proof where the size of the exponent 256 matters, and any exponent of at least 200 would do. So Theorem 1.3 gives

$$p(G, W) \;\leq\; \frac{1}{2} + \frac{20}{\log^{(3)} r} \;\leq\; \frac{1}{2} + \delta, \qquad \text{where } \delta = \frac{20}{\log^{(2)} R}.$$

The second inequality holds because $\log^{(3)} r \geq \log(8 + \log R) \geq \log^{(2)} R$. The graph G has cycles of at least $L - 2$ lengths. So Lemma 2.1(b), with $p = p(G, W)$, and the bounds on t and m give

$$\frac{L-2}{2^r} \;\leq\; \frac{pm+1}{2^t} \;\leq\; \Big(\frac{1}{2} + \delta\Big)\big(\varphi + 2^{-R} + 5R\, 2^{-2R}\big) + 2^{-2R} \;\leq\; \Big(\frac{1}{2} + \delta\Big)\varphi + 2^{2-R}. \qquad \square$$

Corollary 4.3. *There is a constant c such that $\Phi(r) \leq c\, 2^{-\log^* r}$ for all $r \geq 1$.*

Proof. Fix $R_0 \geq 512$ so large that Proposition 4.2 holds for all $R \geq R_0$. Let $R_{j+1} = 2^{256 R_j}$ and $\delta_j = 20/\log^{(2)} R_j$. By Proposition 4.2, the numbers $a_j = 2^j \Phi(R_j)$ satisfy

$$a_{j+1} \;\leq\; (1 + 2\delta_j)\, a_j + 2^{\,j+3-R_j}.$$

For $j \geq 2$ we have $\log^{(2)} R_j = 8 + 256 R_{j-2}$, so $\sum_j \delta_j < \infty$. Also $\sum_j 2^{j-R_j} < \infty$. Hence the numbers a_j are bounded, say $\Phi(R_j) \leq c_1 2^{-j}$.

The numbers R_j grow like a tower of exponentials. To bound $\log^* R_j$, we compare them with an exact tower. Let $U_0 = R_0^2$ and $U_{j+1} = 2^{U_j}$, so that $\log^* U_j = j + \log^* U_0$. Then $R_j \leq \sqrt{U_j}$ for all j. Indeed, if this holds for j, then

$$R_{j+1} = 2^{256 R_j} \;\leq\; 2^{256\sqrt{U_j}} \;\leq\; 2^{U_j/2} = \sqrt{U_{j+1}},$$

because $\sqrt{U_j} \geq R_0 \geq 512$. Hence $\log^* R_j \leq j + \log^* U_0$.

Finally let $r \geq R_0$, and let j be largest with $R_j \leq r$. Then $\Phi(r) \leq \Phi(R_j) \leq c_1 2^{-j}$ and $\log^* r \leq \log^* R_{j+1} \leq j + 1 + \log^* U_0$. Together these give $\Phi(r) \leq c\, 2^{-\log^* r}$ with $c = c_1\, 2^{1+\log^* U_0}$. For $r < R_0$ use $\Phi(r) \leq 1$, and enlarge c if necessary. □

Proof of the lower bound in Theorem 1.4. Let G be a pancyclic graph with n vertices and $n + h$ edges. It is connected, so its rank is $r = h + 1$. By the definition of Φ and by Corollary 4.3,

$$n - 2 \;\leq\; \Phi(r)\, 2^r \;\leq\; c\, 2^{\,r - \log^* r}.$$

We compare $\log^* r$ with $\log^* n$. The trivial bound $n - 2 \leq 2^r - 1$ gives $n \leq 2^{r+1} \leq 2^{2^r}$, hence $\log^* n \leq \log^* r + 2$. Now take logarithms in the display:

$$h = r - 1 \;\geq\; \log(n-2) + \log^* r - \log c - 1 \;\geq\; \log n + \log^* n - \log c - 5. \qquad \square$$

Formal counterpart. In the Lean code Φ is `actualRankTail`. Proposition 4.2 is `ShortProof.recurrence_of_few_component_forest_estimate`, with the same constants. Corollary 4.3 is an instance of `recurrence_logStar_bound`. Lemma 4.1 corresponds to `ShortProof.exists_budgeted_witness`, where the bookkeeping differs in detail.

5 The construction

This section proves the upper bound of Theorem 1.4. The construction follows [11]. It is a chain of beads. Bead i can be passed in two ways, and the two ways differ in length by 2^i. So k beads give 2^k consecutive path lengths for the price of k chords. A few more chords close shorter and shorter parts of the chain, and these supply the short cycles.

Proposition 5.1. *For every $n \geq 3$ there is a pancyclic graph with n vertices and at most $n + \lfloor \log n \rfloor + \log^* n + 1$ edges.*

Proof. For $3 \leq n \leq 6$ take a cycle $u_0 u_1 \cdots u_{n-1}$, with the chord $u_0 u_2$ if $n \geq 4$ and also the chord $u_0 u_3$ if $n = 6$. Now let $n \geq 7$.

The necklace. Let $k \geq 2$ be the largest integer with $2^k + k + 1 \leq n$, so that $n \leq 2^{k+1} + k + 1$. Let H be a cycle of length n, and let $Q \subseteq H$ be a path that is composed of consecutive segments $Q_0, \ldots, Q_{k-1}$, where Q_i has $2^i + 1$ edges and joins v_i to v_{i+1}. Thus Q has $2^k + k - 1 \leq n - 2$ edges. Add the chords $e_i = v_i v_{i+1}$ for $0 \leq i < k$. For $1 \leq j \leq k$ let N_j, the *necklace with j beads*, be the union of $Q_0, \ldots, Q_{j-1}$ and $e_0, \ldots, e_{j-1}$ (Figure 3).

A path from v_0 to v_j in N_j uses either Q_i or e_i for each $i < j$, and it is shorter by 2^i if it uses e_i. So N_j has 2^j such paths, exactly one of each length from j to $2^j + j - 1$. The rest $H - Q$ of the outer cycle is a path from v_k to v_0 with $n - 2^k - k + 1$ edges. Together with N_k it contains a cycle of each length in the interval

$$[\, n - 2^k + 1,\; n \,].$$

Closing chords. For $2 \le j \le k$ the chord $f_j = v_0 v_j$ closes the paths of N_j to cycles of all lengths in the interval

$$I_j = [\, j+1,\ 2^j + j \,].$$

Add the chord f_j for $j = k$, then for $j = \lceil \log k \rceil$, and so on: as long as the last value of j is larger than 2, the next value is $\lceil \log j \rceil$. The last value is 2. If $j' = \lceil \log j \rceil$, then $2^{j'} \ge j$, so the intervals $I_{j'}$ and I_j overlap or are adjacent. Since $I_2 = [3, 6]$, these chords give a cycle of each length in $[\, 3,\ 2^k + k \,]$.

All lengths. The intervals $[\, 3,\ 2^k + k \,]$ and $[\, n - 2^k + 1,\ n \,]$ cover all lengths from 3 to n, except for the length $2^k + k + 1$ in the case $n = 2^{k+1} + k + 1$. In that case add one more chord, which joins v_k to the neighbour of v_0 on $H - Q$. This chord, the edge from that neighbour to v_0, and the path Q form a cycle of length $2^k + k + 1$.

The number of chords. For an integer $j \ge 3$ we have $\log^* \lceil \log j \rceil = \log^* j - 1$, and $\log^* 2 = 1$. So exactly $\log^* k$ chords f_j were added. The number of all chords is therefore at most $k + \log^* k + 1 \le \lfloor \log n \rfloor + \log^* n + 1$. □

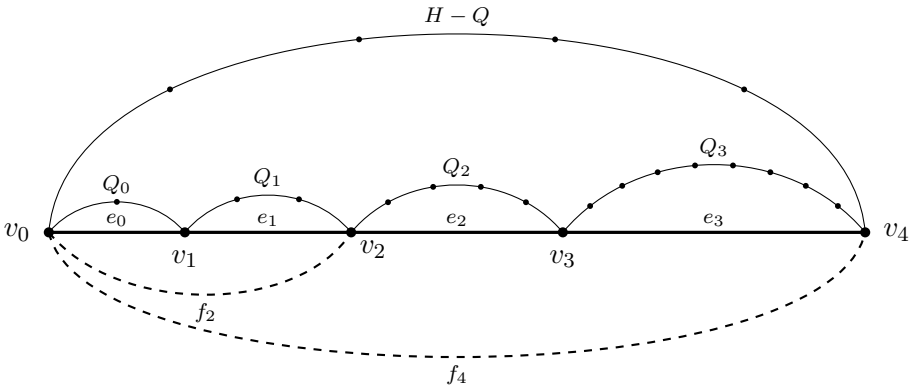


Figure 3: The construction for $n = 24$, where $k = 4$. The segments $Q_0, \ldots, Q_3$ of the outer cycle H have $2, 3, 5, 9$ edges and are doubled by the chords $e_0, \ldots, e_3$. The chord f_4 gives the cycle lengths $5, \ldots, 20$, the rest $H - Q$ of the outer cycle gives $9, \ldots, 24$, and f_2 gives $3, \ldots, 6$.

Formal counterpart. The Lean theorem `Problem1016.upper_bound_integer` states that $h(n) \le \lfloor \log n \rfloor + \log^* n + 1$ for every $n \ge 3$. Its proof formalises the construction of [11]. The proof above describes that construction in our own words. As a check on the wording, we built the graph of this section by computer for all $n \le 139$ and for eight larger values up to 522. Each of these graphs is pancyclic and has at most the stated number of edges (Appendix B).

6 Proof of the forest estimate

This section proves Theorem 1.3. The argument is that of [11]. It has five steps.

1. *Cuts.* For a set A of vertices, consider the event that ξ avoids every edge that leaves A. Its probability is explicit. For two disjoint sets the two events are correlated by the factor $2^{\ell-1}$, where ℓ counts the links between the sets (Lemma 6.2). One link means independence, and two links mean a factor of two.

2. *Reduction.* We pass to a graph Γ with all degrees at most 3, in which W has become a small set P of vertices (Lemma 6.5).

3. *Short cycles.* Step 1 and a second moment bound show that many disjoint short cycles in $\Gamma - P$ already force the estimate (Lemma 6.4). So we may delete a small maximal family of them, and we are left with a graph J of large girth.

4. *Supply.* The graph J has many cycles with between $6\log r$ and $\log r\sqrt{\log\log r}$ edges. Their total weight $\sum_C 2^{-|C|}$ is at least $(\frac{1}{8} - o(1))\ln\log\log r$ (Lemma 6.8). This step uses the non-backtracking matrix.

5. *Second moment.* For such a cycle C, the event that C is a component of ξ has probability at least $2^{-|C|}$. If it occurs, then the part of ξ outside P is not a forest. We weight the indicators of these events so that their means are exactly $2^{-|C|}$. Most pairs of the events are correlated by a factor of at most two.

The factor two in step 5 is the source of the constant $\frac{1}{2}$. Lemma 6.1 turns the bound $\mathbb{E}Z^2 \le 2\mu^2 + K\mu$ into $\mathbb{P}(Z = 0) \le \frac{1}{2} + K/(4\mu)$. Appendix A lists the Lean declarations that correspond to the steps.

In this section graphs may have loops and parallel edges. A loop adds two to the degree of its vertex and is a cycle of length one. Two parallel edges form a cycle of length two. For a set A of vertices of G we write $G[A]$ for the induced subgraph, $G - A$ for $G[V(G) \setminus A]$, and δA for the set of edges with exactly one end in A. A *region* is a nonempty set A such that $G[A]$ is connected. A region is *cyclic* if $G[A]$ contains a cycle. As before ξ is uniform on $\mathcal{Z}(G)$, and $\mathcal{O}_A$ denotes the event $\xi \cap \delta A = \emptyset$.

6.1 Cuts and links

Lemma 6.1. *Let $Z \ge 0$ be a random variable with $\mu = \mathbb{E}Z > 0$ and $\mathbb{E}Z^2 \le 2\mu^2 + K\mu$. Then $\mathbb{P}(Z = 0) \le \frac{1}{2} + K/(4\mu)$.*

Proof. By the Cauchy–Schwarz inequality $\mu^2 \le \mathbb{E}Z^2 \cdot \mathbb{P}(Z > 0)$, so $\mathbb{P}(Z = 0) \le 1 - \mu/(2\mu + K) \le \frac{1}{2} + K/(4\mu)$. □

Let A and B be disjoint regions of a connected graph G. A *link* between them is an edge with one end in A and the other in B, or a component of $G - (A \cup B)$ that is joined by an edge to A and by an edge to B (Figure 4). Let $\ell(A, B)$ be the number of links.

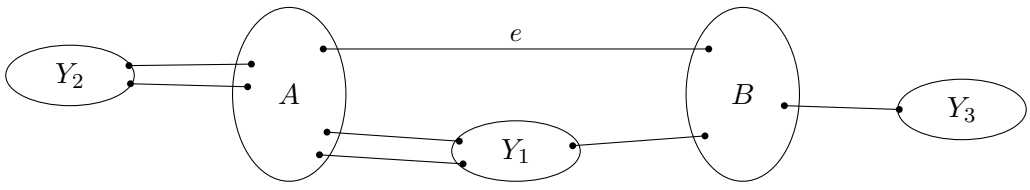


Figure 4: Links between two regions A and B. The edge e and the component Y_1 of $G - (A \cup B)$ are links. The components Y_2 and Y_3 are joined to one region only, so they are not links. Here $\ell(A, B) = 2$. In the proof of Lemma 6.2 this is the case $a = b = k = 1$ and $e(A, B) = 1$.

Lemma 6.2 (Link formula)**.** *Let A and B be disjoint regions of a connected graph G. Then $\mathbb{P}(\mathcal{O}_A) = 2^{\mathrm{comp}(G-A)-|\delta A|} \ge 2^{-|\delta A|}$ and*

$$\mathbb{P}(\mathcal{O}_A \cap \mathcal{O}_B) = 2^{\ell(A,B)-1}\,\mathbb{P}(\mathcal{O}_A)\,\mathbb{P}(\mathcal{O}_B).$$

Conditionally on $\mathcal{O}_A$, the set $\xi \cap E(G[A])$ is uniform on $\mathcal{Z}(G[A])$; conditionally on $\mathcal{O}_A \cap \mathcal{O}_B$, the sets $\xi \cap E(G[A])$ and $\xi \cap E(G[B])$ are independent and uniform on $\mathcal{Z}(G[A])$ and $\mathcal{Z}(G[B])$.

Proof. The even subgraphs that avoid δA are the sums of an even subgraph of $G[A]$ and an even subgraph of $G - A$. So $\mathbb{P}(\mathcal{O}_A) = 2^{\beta(G[A])+\beta(G-A)-\beta(G)}$, and counting edges and vertices shows that the exponent is $\operatorname{comp}(G - A) - |\delta A|$. In the same way

$$\mathbb{P}(\mathcal{O}_A \cap \mathcal{O}_B) = 2^{\beta(G[A])+\beta(G[B])+\beta(G-A-B)-\beta(G)} = 2^{\operatorname{comp}(G-A-B)+1-|\delta A|-|\delta B|+e(A,B)},$$

where $e(A, B)$ is the number of edges between A and B. Let a, b and k be the numbers of components of $G-A-B$ that are joined to A only, to B only, and to both. Then $\operatorname{comp}(G-A-B) = a+b+k$, while $\operatorname{comp}(G-A) = a+1$ and $\operatorname{comp}(G-B) = b+1$, and the formula follows since $\ell(A, B) = k+e(A, B)$. The conditional laws are read off from the two direct sum decompositions. □

So two cut events are independent when there is one link. They are correlated by a factor of exactly two when there are two links. The correlation is stronger only for pairs with at least three links, and the next lemma says that such pairs are rare.

Lemma 6.3 (Exceptional partners)**.** *Let A be a region of a connected graph G, and let $B_1, \dots, B_k$ be pairwise disjoint regions, disjoint from A, with $\ell(A, B_i) \geq 3$ for all i. Then $k \leq |\delta A|$.*

Proof. Contract A to a vertex a and each B_i to a vertex b_i, and delete the loops that arise. This changes neither $d = |\delta A|$ nor the numbers $\ell(A, B_i)$, because the sets B_j with $j \neq i$ are connected and are contracted inside components of $G - (A \cup B_i)$. Let F be a spanning forest of $G - a$, and for every edge $e = au$ at a attach a new leaf τ_e to u. Call the resulting forest F^+ and its d new leaves *terminals*.

Fix i. Every link between a and b_i yields a path of F^+ from b_i to a terminal. For an edge $e = ab_i$ it is the edge $b_i\tau_e$. For a component Y of $G - \{a, b_i\}$ joined to both, choose an edge $e = au$ with $u \in Y$ and take the path of F^+ from b_i to τ_e; after its first vertex it stays in $Y \cup \{\tau_e\}$. Paths that come from different links therefore start with different edges at b_i, and the union of any two of them is a path between two terminals. Let $F^\circ \subseteq F^+$ be the union of all paths between terminals. As $\ell(a, b_i) \geq 3$, the vertex b_i has degree at least 3 in F°. All leaves of F° are terminals, and a forest with at most d leaves has at most $d - 2$ vertices of degree at least 3. Hence $k \leq d - 2$ if $k \geq 1$. □

Lemma 6.4 (Many small cuts)**.** *Let $A_1, \dots, A_k$ be pairwise disjoint cyclic regions of a connected graph G with $|\delta A_i| \leq d$ for all i. Then*

$$\mathbb{P}(\xi \cap E(G[A_i]) \text{ is a forest for every } i) \;\leq\; \frac{1}{2} + \frac{(d+1)\,2^{d-1}}{k}.$$

Proof. Let E_i be the event that $\mathcal{O}_{A_i}$ holds and $\xi \cap E(G[A_i]) \neq \emptyset$. On E_i the set $\xi \cap E(G[A_i])$ is a nonempty even subgraph, hence not a forest, so it suffices to bound $\mathbb{P}(Z = 0)$ for $Z = \sum_i \mathbf{1}_{E_i}$. By Lemma 6.2, $p_i := \mathbb{P}(E_i) = \mathbb{P}(\mathcal{O}_{A_i})(1 - 2^{-\beta(G[A_i])}) \geq 2^{-d-1}$ and $\mathbb{P}(E_i \cap E_j) = 2^{\ell(A_i,A_j)-1} p_i p_j$ for $i \neq j$. If $\ell(A_i, A_j) \leq 2$ this is at most $2p_ip_j$. Otherwise we use $\mathbb{P}(E_i \cap E_j) \leq p_i$; by Lemma 6.3 this happens for at most d indices j, for each i. Hence $\mathbb{E}Z^2 \leq \mu + 2\mu^2 + d\mu$ with $\mu = \mathbb{E}Z \geq k\,2^{-d-1}$, and Lemma 6.1 gives the claim. □

6.2 Reduction to a subcubic graph

Lemma 6.5. *Let G and W be as in Theorem 1.3, with $m = |E(W)|$ and $c = \operatorname{comp}(W)$. There are a connected multigraph Γ with $\beta(\Gamma) = r$ and all degrees at most 3, and a set $P \subseteq V(\Gamma)$, such that*

(i) *every vertex outside P has degree 3, $|P| \le 2m$, and $\Gamma[P]$ has at most c components;*

(ii) $p(G,W) \le \mathbb{P}\big(\xi_\Gamma \cap E(\Gamma - P)$ *is a forest*$\big)$*, where ξ_Γ is uniform on $\mathcal{Z}(\Gamma)$.*

In particular $2r - 2 \le |V(\Gamma)| \le 2r + 2m$.

Proof. Splitting. Replace every vertex v of degree $d \ge 4$ by a path $v^1 v^2 \cdots v^{d-2}$. Distribute the d edges at v along this path in some order: two to v^1, one to each of $v^2, \dots, v^{d-3}$, and two to v^{d-2}. If $v \in V(W)$, let the edges of W at v come first in this order, so that they end in an initial segment $v^1, \dots, v^i$ of the path with $i < \deg_W(v)$. Let G' be the new graph; its degrees are at most 3. Let P consist of these initial segments and of the vertices of W that were not split. Then $|P| \le \sum_v \deg_W(v) = 2m$, every edge of W joins two vertices of P, and $G'[P]$ has at most c components.

Contracting the new path edges maps $\mathcal{Z}(G')$ bijectively onto $\mathcal{Z}(G)$, so we may take $\xi_G = \xi_{G'} \cap E(G)$. Every edge of $G' - P$ is a new path edge or an edge of $E(G) \setminus E(W)$. If $\xi_G \setminus E(W)$ is a forest, then so is its union with the set of all new path edges, because contracting these edges, which form a forest, gives back $\xi_G \setminus E(W)$. Hence $p(G,W) \le \mathbb{P}\big(\xi_{G'} \cap E(G' - P)$ is a forest$\big)$.

Cleaning. As long as possible, delete a vertex of degree one outside P, or replace a vertex of degree two outside P and its two edges by a single edge. An even subgraph avoids a pendant edge and contains both or none of the two edges at a vertex of degree two, so these operations induce bijections of the cycle spaces. Let Γ be the final multigraph. Each edge of $\Gamma - P$ is a path of G' all of whose vertices lie outside P; so if ξ_Γ contains a cycle of $\Gamma - P$, then the corresponding $\xi_{G'}$ contains a cycle of $G' - P$. This proves (ii).

Edges with both ends in P are never removed, so the vertices of P keep degree at least two, and the number of components of the graph induced on P does not increase. The vertices outside P now have degree exactly 3. Finally $3|V(\Gamma)| - |P| \le 2|E(\Gamma)| \le 3|V(\Gamma)|$ and $|E(\Gamma)| = |V(\Gamma)| + r - 1$ give the bounds on $|V(\Gamma)|$. □

6.3 Parameters, and removing the short cycles

From now on Γ and P are as in Lemma 6.5, $\xi = \xi_\Gamma$, and r is large. All $o(\cdot)$ and $O(\cdot)$ refer to $r \to \infty$ and are uniform in G and W. Put

$$x = \log r, \qquad y = \log x, \qquad z = \log y,$$
$$D = \lfloor x/10 \rfloor, \qquad L = \lfloor x\sqrt{y} \rfloor, \qquad s = \lfloor (D-1)/2 \rfloor, \qquad \varepsilon = 20/z.$$

These numbers have the following roles. We delete cycles of length at most D, so the graph that remains has girth greater than D. We count cycles of length at most L. A non-backtracking walk of length s in the remaining graph is determined by its two ends. The number ε is the error term of the theorem. These letters keep their meaning until the end of the section.

Recall that $|P| \le 2\sqrt{r}$ and $c \le x/100$. Let $\mathcal{F}$ be the event that $\xi \cap E(\Gamma - P)$ is a forest. We assume that

$$\mathbb{P}(\mathcal{F}) > \tfrac{1}{2} + \varepsilon \tag{6.1}$$

and derive a contradiction. By Lemma 6.5(ii) this proves Theorem 1.3.

Let $\mathcal{Q}$ be a maximal family of pairwise disjoint cycles of $\Gamma - P$ of length at most D, loops and pairs of parallel edges included, and let S be the union of their vertex sets. For $Q \in \mathcal{Q}$ the set $V(Q)$ is a cyclic region of Γ with $|\delta V(Q)| \le |Q| \le D$, because its vertices have degree 3. On $\mathcal{F}$ the set $\xi \cap E(\Gamma[V(Q)])$ is a forest for every Q, so Lemma 6.4 and (6.1) give $|\mathcal{Q}| < (D+1)2^{D-1}/\varepsilon \le \sqrt{r}$. Hence $|S| \le D\sqrt{r}$.

Let $J = \Gamma - (P \cup S)$. By the maximality of $\mathcal{Q}$, J is a simple graph of girth greater than D. It has $v = 2r + O(\sqrt{r}\log r)$ vertices, each of degree 3 in Γ, so its *deficit* satisfies

$$d_J := \sum_{u \in V(J)} \bigl(3 - \deg_J u\bigr) \;\le\; 3\bigl(|P| + |S|\bigr) = O(\sqrt{r}\log r). \tag{6.2}$$

6.4 Cycles of moderate length

A walk $u_0 u_1 \cdots u_\ell$ is *non-backtracking* if $u_{i+1} \neq u_{i-1}$ for $0 < i < \ell$. Let $N_\ell(u, w)$ be the number of non-backtracking walks of length ℓ from u to w in J.

Lemma 6.6. *If $\ell > s$, then $N_\ell(u, w) \le 3 \cdot 2^{\ell - s - 1}$ for all u, w. Consequently, for every vertex u of J, the cycles C of J through u with $|C| \le L$ satisfy $\sum_C 2^{-|C|} \le L\, 2^{-s}$.*

Proof. There are at most $3 \cdot 2^{\ell-s-1}$ ways to choose the first $\ell - s$ steps. The last s steps form a non-backtracking walk of length s between two given vertices. Such a walk is a path, because s is less than the girth, and there is at most one such path, because two of them would give a cycle of length at most $2s < D$. For the second statement, a cycle of length ℓ through u can be traversed from u in two directions, so there are at most $\frac{1}{2} N_\ell(u, u)$ of them, and $\ell > D > s$. □

Let B be the non-backtracking matrix of J. Its rows and columns are indexed by the directed edges $(i \to j)$ of J, two for each edge. The entry in row $(i \to j)$ and column $(j' \to k)$ is 1 if $j' = j$ and $k \neq i$, and it is 0 otherwise.

Lemma 6.7. *For every even $\ell \ge 2$,*

$$\operatorname{tr} B^\ell \;\ge\; \bigl(2 - d_J/v\bigr)^\ell - 3v\, 2^{\ell/2}.$$

Proof. Let A be the adjacency matrix of J and Δ the diagonal matrix of its degrees, and put $M(\lambda) = \lambda^2 I - \lambda A + \Delta - I$.

(1) *If $M(\lambda) g = 0$ for a real $\lambda > 1$ and a vector $g \neq 0$, then λ is an eigenvalue of B.* A direct computation shows that $f(i \to j) = \lambda g(j) - g(i)$ satisfies $Bf = \lambda f$; and $f = 0$ would give $g(i) = \lambda g(j) = \lambda^2 g(i)$ along every edge, hence $g = 0$ (at an isolated vertex, $M(\lambda) g = 0$ already says $(\lambda^2 - 1) g = 0$).

(2) *If $Bf = \lambda f$ with $f \neq 0$ and $\lambda^2 \neq 1$, then $M(\lambda) g = 0$ for $g(j) = \sum_{k \sim j} f(j \to k)$, and $g \neq 0$.* The eigenvalue equation at $(i \to j)$ and at $(j \to i)$ reads $\lambda f(i \to j) = g(j) - f(j \to i)$ and $\lambda f(j \to i) = g(i) - f(i \to j)$. Eliminating $f(j \to i)$ gives $(\lambda^2 - 1) f(i \to j) = \lambda g(j) - g(i)$. This shows that $g \neq 0$. Summing it over the neighbours j of i gives $M(\lambda) g = 0$.

(3) *B has a real eigenvalue $\lambda_1 \ge 2 - d_J/v$.* The matrix $M(2) = 2(\Delta - A) + (3I - \Delta)$ is positive semidefinite, while $\mathbf{1}^\top M(\lambda) \mathbf{1} = (\lambda - 1)(v\lambda - 2v + d_J)$ vanishes at $\lambda^* = 2 - d_J/v > 1$. The smallest eigenvalue of the symmetric matrix $M(\lambda)$ is continuous in λ, at most 0 at λ^* and at least 0 at 2. So $M(\lambda_1)$ is singular for some $\lambda_1 \in [\lambda^*, 2]$, and (1) applies.

(4) *Every non-real eigenvalue λ of B satisfies $|\lambda| \le \sqrt{2}$.* With g as in (2), $g^* M(\lambda) g = 0$ is a quadratic equation $a\lambda^2 - b\lambda + c' = 0$ with real coefficients $a = g^* g > 0$, $b = g^* A g$ and $c' = g^*(\Delta - I) g \le 2a$. Its roots are λ and $\bar{\lambda}$, so $|\lambda|^2 = c'/a \le 2$.

Now $\operatorname{tr} B^\ell$ is the sum of the ℓth powers of the $2|E(J)| \le 3v$ eigenvalues of B. For even ℓ the real eigenvalues contribute nonnegative terms, one of which is at least $(2 - d_J/v)^\ell$, and the non-real ones contribute at least $-3v\, 2^{\ell/2}$ in total. □

Facts (2) and (4) are part of the Ihara–Bass theory [3, 14]. The test vector $\mathbf{1}$ in (3) is from [11]. It makes the argument work without any assumption of regularity or expansion.

Lemma 6.8 (Cycle weight)**.** *Let $\mathcal{C}$ be the set of cycles of J of length at most L. Their total weight satisfies*

$$\mu := \sum_{C\in\mathcal{C}} 2^{-|C|} \;\ge\; (\tfrac{1}{8} - o(1)) \ln y.$$

Proof. $\operatorname{tr} B^\ell$ is the number of closed walks $u_0u_1\cdots u_\ell = u_0$ in J with $u_{i+1} \neq u_{i-1}$ for all i modulo ℓ. A cycle of length ℓ gives 2ℓ of them; let n_ℓ be the number of the others. Then

$$T := \sum_{\ell\le L} \frac{\operatorname{tr} B^\ell}{2\ell\, 2^\ell} = \mu + \sum_{\ell\le L} \frac{n_\ell}{2\ell\, 2^\ell}.$$

Consider a walk counted by n_ℓ, and let j be the least index with $u_j \in \{u_0,\dots,u_{j-1}\}$, say $u_j = u_i$. Then $j < \ell$, the walk $C = u_iu_{i+1}\cdots u_j$ is a cycle, of length q say, and $u_ju_{j+1}\cdots u_\ell u_1\cdots u_i$ is a non-backtracking closed walk of length $\ell - q > 0$ from u_i to itself. It contains a cycle, so $\ell - q > D > s$. The original walk is determined by C, by the vertex u_i and a direction on C ($2q$ choices), by the complementary walk (at most $3\cdot 2^{\ell-q-s-1}$ choices, by Lemma 6.6), and by the position of u_0 on the latter (at most ℓ choices). Hence

$$\sum_{\ell\le L} \frac{n_\ell}{2\ell\, 2^\ell} \;\le\; \sum_{\ell\le L} \sum_{C\in\mathcal{C}} \tfrac{3}{2}\,|C|\, 2^{-|C|-s} \;\le\; \tfrac{3}{2}\, L^2\, 2^{-s}\, \mu = o(\mu),$$

because $2^{-s} \le 8\, r^{-1/20}$ and $L^2 \le x^2 y$.

On the other hand, for even ℓ with $6x \le \ell \le L$, Lemma 6.7 and (6.2) give

$$2^{-\ell} \operatorname{tr} B^\ell \;\ge\; \Big(1 - \frac{d_J}{2v}\Big)^\ell - 3v\, 2^{-\ell/2} \;\ge\; 1 - \frac{L\, d_J}{2v} - 3v\, r^{-3} = 1 - o(1).$$

All terms of T are nonnegative, so $T \ge (1-o(1)) \sum 1/(2\ell)$ over the even ℓ in $[6x, L]$, and this sum is $(\frac{1}{4} - o(1)) \ln(L/6x) = (\frac{1}{8} - o(1)) \ln y$. Therefore $\mu = (1-o(1))T \ge (\frac{1}{8} - o(1)) \ln y$. □

6.5 The second moment

For $C \in \mathcal{C}$ let I_C be the event that C is a component of ξ. As the vertices of C have degree 3 in Γ, this is the event $E(C) \subseteq \xi$. It is also the event that $\mathcal{O}_{V(C)}$ holds and $\xi \cap E(\Gamma[V(C)]) = E(C)$. Put

$$a_C = 2^{-|C|}, \qquad p_C = \mathbb{P}(I_C), \qquad X_C = \frac{a_C}{p_C}\, \mathbf{1}_{I_C}, \qquad Z = \sum_{C\in\mathcal{C}} X_C.$$

The factor a_C/p_C fixes the mean of X_C at the combinatorial weight a_C, whatever the probability p_C is. We have no upper bound for p_C, and with this normalisation we need none. The event I_C is given by $|C|$ linear conditions on $\xi \in \mathcal{Z}(\Gamma)$, which are satisfied by $\xi = C$. So $p_C \ge a_C$. Hence $0 \le X_C \le 1$, and $\mathbb{E}Z = \mu$ with μ as in Lemma 6.8. On $\mathcal{F}$ no cycle of $J \subseteq \Gamma - P$ is contained in ξ, so $Z = 0$, and by (6.1)

$$\mathbb{P}(Z = 0) > \tfrac{1}{2} + \varepsilon. \tag{6.3}$$

Let $C \neq C'$ be in $\mathcal{C}$. If they share a vertex, then $I_C \cap I_{C'} = \emptyset$. If they are disjoint, Lemma 6.2, applied to the regions $V(C)$ and $V(C')$ of Γ, gives $\mathbb{P}(I_C \cap I_{C'}) = 2^{\ell(C,C')-1} p_C p_{C'}$ and therefore

$$\mathbb{E}(X_C X_{C'}) = a_C a_{C'}\, 2^{\ell(C,C')-1}, \tag{6.4}$$

where $\ell(C,C')$ is the number of links between $V(C)$ and $V(C')$ in Γ.

Lemma 6.9 (Few links)**.** *$\ell(C, C') \le c + o(x)$ for all disjoint $C, C' \in \mathcal{C}$.*

Proof. Let $U = V(C) \cup V(C')$. Every vertex of U lies on exactly one edge outside its own cycle, so at most $2L$ edges leave U. At most c components of $\Gamma - U$ meet P, because $P \cap U = \emptyset$ and so each component of $\Gamma[P]$ lies in one of them. Let Y run over the other components and let d_Y be the number of edges between Y and U; then $\sum_Y d_Y \le 2L$. Put $t_0 = \lfloor y/2 \rfloor$.

Large cuts. At most $2L/t_0$ components have $d_Y > t_0$.

Small cuts, cyclic. The components Y with $d_Y \le t_0$ that are cyclic are pairwise disjoint cyclic regions inside $\Gamma - P$ with $|\delta Y| \le t_0$. On $\mathcal{F}$ each $\xi \cap E(\Gamma[Y])$ is a forest, so by Lemma 6.4 and (6.1) there are fewer than $(t_0 + 1)2^{t_0 - 1}/\varepsilon$ of them.

Small cuts, trees. Let Y induce a tree, with $d_Y \le t_0$. No $Q \in \mathcal{Q}$ meets Y, since Q is a connected subgraph of $\Gamma - U$ and would lie in Y. Hence $Y \subseteq V(J)$. As all degrees are 3, we have $3|Y| = 2(|Y| - 1) + d_Y$, that is, $|Y| = d_Y - 2$. So if Y is joined to both cycles, there is a path R from C to C' through Y with at most t_0 edges. For a link that is an edge, let R be that edge. The paths R obtained in this way lie in J and are pairwise disjoint. Divide each of C and C' into at most $8L/D$ blocks of at most $D/4$ consecutive vertices. If two of the paths had their ends in the same block of C and in the same block of C', they would form, together with two arcs inside these blocks, a cycle of J of length less than $2t_0 + D/2 \le D$. So there are at most $(8L/D)^2$ such paths.

Altogether

$$\ell(C, C') \;\le\; c + \frac{2L}{t_0} + \frac{(t_0 + 1)2^{t_0 - 1}}{\varepsilon} + \Big(\frac{8L}{D}\Big)^2 = c + O\Big(\frac{x}{\sqrt{y}}\Big) + O(yz\sqrt{x}) + O(y) = c + o(x). \quad \square$$

Proof of Theorem 1.3. Call a disjoint pair $C, C' \in \mathcal{C}$ *exceptional* if $\ell(C, C') \ge 3$. Fix C, and let $\mathcal{E}$ be a maximal family of pairwise disjoint exceptional partners of C. By Lemma 6.3, $|\mathcal{E}| \le |\delta V(C)| \le L$, so the union U_C of their vertex sets has at most L^2 vertices, and every exceptional partner of C meets U_C. By Lemma 6.6, the sum of $a_{C'}$ over all $C' \in \mathcal{C}$ that meet U_C is at most $L^3 2^{-s}$. With (6.4) and Lemma 6.9,

$$\sum_{C' \text{ exceptional}} \mathbb{E}(X_C X_{C'}) \;\le\; a_C \, 2^{c + o(x)} L^3 \, 2^{-s} = a_C \, 2^{c - x/20 + o(x)} = o(a_C),$$

because $c \le x/100$.

For the other disjoint pairs $\mathbb{E}(X_C X_{C'}) \le 2a_C a_{C'}$. Intersecting pairs contribute nothing, and $\mathbb{E}X_C^2 \le \mathbb{E}X_C = a_C$. Summing over C we get $\mathbb{E}Z^2 \le 2\mu^2 + (1 + o(1))\mu$. By Lemmas 6.1 and 6.8,

$$\mathbb{P}(Z = 0) \;\le\; \frac{1}{2} + \frac{1 + o(1)}{4\mu} \;\le\; \frac{1}{2} + \frac{2 + o(1)}{\ln y} \;<\; \frac{1}{2} + \frac{20}{z}$$

for large r, since $\ln y = z \ln 2$. This contradicts (6.3). $\square$

Remark 6.10. The assumption $|E(W)| \le \sqrt{r}$ entered only through $|P| \le 2\sqrt{r}$ in (6.2), where all that is needed is $L \, d_J = o(v)$; the proof therefore works unchanged for $|E(W)| \le r/\log^2 r$. Similarly the constant 20 can be replaced by any constant larger than $2/\ln 2$. We have kept the statement of [11], which is the one that has been formalised.

7 The sharp bound: the parts of the proof

Sections 7 to 13 prove Theorem 1.1. This section states the three parts of the proof and derives the theorem from them. Sections 8 to 11 prove the first part, Theorem 1.5. Section 12 describes the other two parts, whose proofs are in the supplement. None of these sections is formalised as a whole. Table 3 lists the steps that are.

7.1 Conventions

As in Section 6, graphs may have loops and parallel edges. A loop adds two to the degree of its vertex. A loop is a cycle of length one, and two parallel edges form a cycle of length two. A multigraph is *cubic* if every vertex has degree three, and *subcubic* if every vertex has degree at most three. The *girth* of a multigraph is the length of a shortest cycle; a forest has infinite girth. We write $c(G)$ for the number of cycles of G.

If we replace an edge by a path, the rank and the number of cycles do not change. So $\psi(r)$ is also the largest number of cycles of a multigraph of rank r.

A *cyclic cut* of G is a set δX of edges such that $G[X]$ and $G - X$ each contain a cycle. Its *size* is $|\delta X|$.

The two auxiliary theorems are stated for cubic multigraphs without loops.

Theorem 7.1 (Small cyclic cuts; Astra, supplement [19])**.** *There is an absolute constant K_4 such that every connected cubic multigraph without loops, of rank r, that has a cyclic cut of size at most four satisfies*

$$c(G) \;\le\; 2^{r-1} + K_4\, r\, 2^{r/2}.$$

Theorem 7.2 (Bounded girth; Astra, supplement [20])**.** *For every integer $g_0 \ge 2$ there is $R(g_0)$ with the following property. Let G be a connected cubic multigraph without loops, of rank $r \ge R(g_0)$ and girth at most g_0, that has no cyclic cut of size at most four. Then $c(G) < 2^{r-1}$.*

The graphs of Horak and Kim have cyclic cuts of size four, and they have more than 2^{r-1} cycles. So the second term of Theorem 1.1 comes from Theorem 7.1 alone. In the other two cases the proof gives no excess over 2^{r-1} beyond the excess at smaller rank.

7.2 Cubic graphs

Lemma 7.3. *For every $r \ge 2$ there is a connected cubic multigraph of rank r with exactly $\psi(r)$ cycles.*

Proof. The numbers of cycles of the graphs of rank r are integers below 2^r, so some graph G_0 attains $\psi(r)$. If G_0 is not connected, identify a vertex of one component with a vertex of another. This changes neither the rank nor the set of cycles. Now delete a vertex of degree one, or replace a vertex of degree two and its two edges by one edge, as long as this is possible. These steps keep the rank and the number of cycles. They may produce loops and parallel edges. The result is a connected multigraph with all degrees at least three, because $r \ge 2$.

Let v be a vertex of degree $d \ge 4$. Replace v by a path $v^1 v^2 \cdots v^{d-2}$, and distribute the d edge ends at v along this path: two at v^1, one at each of $v^2, \ldots, v^{d-3}$, and two at v^{d-2}. The new multigraph has $d-3$ more vertices and $d-3$ more edges, so its rank is r. A cycle through v uses two edge ends at v. Together with the part of the path between these two ends it is a cycle of the new multigraph. Different cycles give different cycles, and the cycles that avoid v remain. So the number of cycles does not decrease, and it cannot increase beyond $\psi(r)$. Repeat this for every vertex of degree at least four. □

Entringer and Slater proved the same statement with simple cubic graphs for $r \ge 3$ [6]. We do not need it in that form.

7.3 Proof of Theorem 1.1

The lower bound for $\psi(r)$ is (1.1); Section 13 recalls the graphs. For the upper bound, let D_* and R be as in Theorem 1.5, let K_4 be as in Theorem 7.1, and let $R(\cdot)$ be as in Theorem 7.2. Choose $R_0 \geq \max\{R,\, R(D_*),\, 9\}$ so large that $2\log(2r) \leq 3\log r$ for $r \geq R_0$. Put

$$K = \max\{K_4,\ 2^{R_0/2}\}.$$

We prove by induction on r that $\psi(r) \leq 2^{r-1} + Kr2^{r/2}$. For $1 \leq r < R_0$ this holds because $\psi(r) < 2^r \leq 2^{r-1} + 2^{R_0/2}2^{r/2}$. Let $r \geq R_0$, and let G be a connected cubic multigraph of rank r with $\psi(r)$ cycles (Lemma 7.3). There are four cases.

Case 1: G has a loop e. The loop is the only cycle through e, and $G - e$ is connected of rank $r-1$. So $c(G) = c(G-e) + 1 \leq 2^{r-2} + K(r-1)2^{(r-1)/2} + 1 \leq 2^{r-1} + Kr2^{r/2}$.

Case 2: G has no loop, and it has a cyclic cut of size at most four. Then Theorem 7.1 gives the bound, because $K \geq K_4$.

Case 3: G has no loop and no cyclic cut of size at most four, and its girth is less than D_*. Then $c(G) < 2^{r-1}$ by Theorem 7.2 with $g_0 = D_*$.

Case 4: G has no loop and its girth is at least D_*. Let W be a shortest cycle and let g be its length. A cubic multigraph of girth g has at least $2^{g/2}$ vertices, and G has $2r-2$ vertices. So $g \leq 2\log(2r) \leq 3\log r$. The set $P = V(W)$ is connected, and $G - P$ has no cycle of length less than D_*. So $q(G,P) \leq \frac{1}{2}$ by Theorem 1.5. Let e be an edge of W. By Lemma 2.3 and the induction hypothesis for the connected multigraph $G - e$,

$$c(G) \ \leq\ c(G-e) + 2^{r-2} \ \leq\ 2^{r-1} + K(r-1)2^{(r-1)/2} \ \leq\ 2^{r-1} + Kr2^{r/2}.$$

This completes the induction. A graph of rank r that is not connected has at most $\psi(r)$ cycles by definition. □

Remark 7.4. In Case 3 the girth is at least five when $r \geq 9$, although the proof does not need this. Let C be a cycle of length at most four in a cubic multigraph without loops, and put $X = V(C)$. At most eight edges have an end in X, and at most four of them leave X. So $G - X$ has rank at least $r - 8$. If $r \geq 9$, then δX is a cyclic cut of size at most four.

Remark 7.5. The multigraph $G - e$ in Case 4 is not cubic. This is why the induction is on $\psi(r)$ and not on a class of cubic graphs.

8 Pieces with a small boundary

This section proves that two disjoint pieces of a cubic graph, each large, with a small boundary and a large girth, force $q(G,P) \leq \frac{9}{25}$ (Proposition 8.9). The only input is Theorem 1.3, in the following form.

Proposition 8.1 (Forest input)**.** *There is an integer $R_* \geq 2^{100}$ with the following property. Let H be a connected multigraph of rank $\rho \geq R_*$, and let W be a cycle of H with at least three and at most $\rho/\log^2\rho$ edges. Then*

$$\mathbb{P}(\xi_H \setminus E(W) \text{ is a forest}) \ \leq\ \frac{11}{20}.$$

Proof. Replace every edge of H outside W by a path with three edges. The new graph H' is simple and connected, it contains W, and it has rank ρ. The even subgraphs of H' correspond to those of H, and one has a forest outside W if and only if the other has. Apply Theorem 1.3 to H' and W, in the range of Remark 6.10: W is one cycle with at most $\rho/\log^2\rho$ edges. Choose R_* so large that $\rho \geq R_*$ is in the range of the theorem, $\log\rho \geq 100$, and $20/\log^{(3)}\rho \leq \frac{1}{20}$. □

Status of the input. Theorem 1.3 is formalised for $|E(W)| \le \sqrt{r}$. The range $|E(W)| \le r/\log^2 r$ is Remark 6.10, which is not formalised. Lemma 8.3 below needs the larger range.

8.1 Localisation

Lemma 8.2 (Localisation)**.** *Let G be a connected multigraph, and let X be a nonempty proper set of vertices such that $G - X$ is connected. Put $d = |\delta X|$. Form G_X from $G[X]$ as follows. For each edge of δX add a new vertex and join it to the end of that edge in X. Join the new vertices by a cycle W_X through all of them; if $d < 3$, first add $3 - d$ further new vertices on that cycle. Then*

(a) *G_X is connected, and $\beta(G_X) = \beta(G[X]) + d - \mathrm{comp}(G[X]) + 1$;*

(b) *if every vertex of X has degree three in G, then $\beta(G_X) = (|X| + d)/2 + 1$;*

(c) *the set $\xi \cap E(G[X])$ has the same distribution for ξ uniform on $\mathcal{Z}(G)$ and for ξ uniform on $\mathcal{Z}(G_X)$;*

(d) *an even subgraph ξ of G_X has a forest outside $E(W_X)$ if and only if $\xi \cap E(G[X])$ is a forest.*

Proof. (a) Every component of $G[X]$ has an edge to $G - X$, because G is connected. So the cycle W_X joins all components. Without the extra vertices, G_X has d more vertices and $2d$ more edges than $G[X]$, and it is connected. Extra vertices on the cycle add as many edges as vertices.

(b) Count the edge ends at X: $3|X| = 2|E(G[X])| + d$. With (a) this gives the formula.

(c) Contract the connected set $V(G) \setminus X$ of G to one vertex, and delete the loops that arise. Call the result G/X. An even subgraph of G restricts to an even subgraph of G/X. Every even subgraph of G/X arises in this way: the missing edges inside the contracted set can be chosen in a spanning tree of that set. So the restriction is a surjective linear map $\mathcal{Z}(G) \to \mathcal{Z}(G/X)$, and it sends the uniform distribution to the uniform distribution. Now contract W_X in G_X. The result is the same multigraph G/X, with the new edges in the place of the edges of δX. So both distributions are the image of the uniform distribution on $\mathcal{Z}(G/X)$.

(d) The edges of G_X outside W_X are the edges of $G[X]$ and the d new edges at X. In $G_X - E(W_X)$ each of these new edges has an end of degree one. So it lies on no cycle of $G_X - E(W_X)$. □

A version of this lemma for connected X was the starting point. The version above, for a set X with a connected complement, is due to Astra.

8.2 Thin markers force thin layers

Lemma 8.3 (Layers)**.** *Put $B = 2R_* + 2^{100} + 1$. Let G be a connected subcubic multigraph, and let P be a nonempty set of vertices such that $G[P]$ is connected and every vertex outside P has degree three. Fix a vertex o, and call the set of vertices at distance j from o the* layer *j. Suppose that P has at most $k \ge 1$ vertices in each layer, and that $q(G, P) \ge \frac{3}{5}$. Then every layer has at most $B(k+1)^2$ vertices.*

Proof. Put $\Phi(m) = m/\log^2 m$ and $\Theta = \max\{2R_*, 2^{100}(k+1)^2\}$.

Step 1: no large set outside P has a small boundary. We claim that no set $X \subseteq V(G) \setminus P$ has $|X| = m \ge \Theta$ and $|\delta X| \le \Phi(m)/4$. Suppose that X is such a set. Let C_0 be the component of $G - X$ that contains P, and put $X^+ = V(G) \setminus V(C_0)$. Then $X \subseteq X^+$, the set X^+ avoids P, and $G - X^+ = C_0$ is connected. Different components of $G - X$ have no edge between them, so

$\delta X^+ \subseteq \delta X$. Every vertex of X^+ has degree three. Let $d^+ = |\delta X^+| \geq 1$ and apply Lemma 8.2 to X^+. The rank of G_{X^+} is

$$\rho = \frac{|X^+| + d^+}{2} + 1 \;\geq\; \frac{m}{2} \;\geq\; R_*.$$

The cycle W_{X^+} has $\max\{d^+, 3\}$ edges. The function Φ is increasing on $[8, \infty)$, and $\Phi(m)/4 \leq \Phi(m/2)$. So $d^+ \leq \Phi(m)/4 \leq \Phi(m/2) \leq \Phi(\rho)$, and also $3 \leq \Phi(\rho)$. By Proposition 8.1 and Lemma 8.2(c),(d),

$$\mathbb{P}(\xi_G \cap E(G[X^+]) \text{ is a forest}) \;\leq\; \frac{11}{20}.$$

But $G[X^+]$ is a subgraph of $G - P$, so this probability is at least $q(G, P) \geq \frac{3}{5}$. This contradiction proves the claim.

Step 2: a window of layers. Let m_j be the number of vertices of layer j outside P, with $m_j = 0$ outside the range of the layers. For $i \leq i'$ let $X(i, i')$ be the set of vertices outside P in the layers $i, \ldots, i'$. An edge joins vertices of the same layer or of consecutive layers. An edge that leaves $X(i, i')$ ends outside P in layer $i-1$ or $i'+1$, or it ends in P in one of the layers $i-1, \ldots, i'+1$. So

$$|\delta X(i, i')| \;\leq\; 3m_{i-1} + 3m_{i'+1} + 3k(i' - i + 3). \tag{8.1}$$

Suppose that $m_j \geq \Theta$ for some j. Start with the window $i = i' = j$. As long as a layer next to the window has more than $m/(48 \log^2 m)$ vertices outside P, where $m = |X(i, i')|$ is the present size, add that layer to the window. The graph is finite, so this stops. Let μ be the final size and $\lambda = \log \mu \geq 100$. Each step multiplies the size by more than $1 + 1/(48\lambda^2)$. So the number of steps is at most

$$\frac{\ln \mu}{\ln(1 + 1/(48\lambda^2))} \;\leq\; 96\lambda^2 \ln \mu \;<\; 67\lambda^3.$$

At the end both layers next to the window have at most $\mu/(48\lambda^2)$ vertices outside P. By (8.1),

$$|\delta X(i, i')| \;\leq\; \frac{\mu}{8\lambda^2} + 3k(67\lambda^3 + 3) \;\leq\; \frac{\mu}{4\lambda^2} = \frac{\Phi(\mu)}{4}.$$

The second inequality says $24k\lambda^2(67\lambda^3 + 3) \leq 2^\lambda$. It holds because $k \leq 2^{\lambda/2-50}$, by $\mu \geq \Theta \geq 2^{100}(k+1)^2$, and because $24\lambda^2(67\lambda^3 + 3) \leq 2000\lambda^5 \leq 2^{\lambda/2+50}$ for $\lambda \geq 100$. So the final window contradicts Step 1.

Hence every layer has fewer than Θ vertices outside P and at most k vertices in P. Their number is at most $\Theta + k \leq B(k+1)^2$. □

8.3 Poles and their forest proportions

A *pole* is a connected multigraph in which every vertex has degree two or three. Its vertices of degree two are its *ports*. Let N be a pole and let T be a set of ports of even size. A *T-join* of N is a set of edges in which the vertices of T have odd degree and all other vertices have even degree. There are $2^{\beta(N)}$ of them. The *forest proportion* of T is the proportion of the T-joins that are forests. For $T = \emptyset$ the T-joins are the even subgraphs, only the empty one is a forest, and the forest proportion is $2^{-\beta(N)}$.

Lemma 8.4 (A marker in a pole)**.** *Let N be a pole with $k \geq 2$ ports. Suppose that some set T of ports has forest proportion at least $\frac{3}{5}$. Then, for every port o, each layer of N with respect to o has at most $B(k+1)^2$ vertices.*

Proof. Take a breadth-first tree of N from o, and let P be the union of its paths from o to the other ports. Then $N[P]$ is connected, P contains all vertices of degree two, and P has at most k vertices in each layer, because a path of the tree from o has one vertex in each layer that it meets. The tree paths inside P contain a T-join F_0. The map $J \mapsto J + F_0$ is a bijection from the T-joins of N to the even subgraphs of N, and $J + F_0$ has the same edges in $N - P$ as J. So if J is a forest, then $J + F_0$ is a forest in $N - P$. Hence $q(N, P) \geq \frac{3}{5}$, and Lemma 8.3 applies to N and P. □

Lemma 8.5 (Girth and rank of a pole)**.** *Let N be a pole of rank $h \geq 2$ with k ports and girth at least D. Put $t = \lceil \log(8h) \rceil$. Then*

$$D \;\leq\; 4t\Big(1 + \frac{k}{3(h-1)}\Big).$$

Proof. Replace every vertex of degree two and its two edges by one edge, and give each edge of the result the length of the path of N that it replaces. The result N_0 is a connected cubic multigraph with $n_0 = 2h - 2$ vertices and $m_0 = 3h - 3$ edges, and the lengths of its edges add up to $m_0 + k$. A cycle of N_0 comes from a cycle of N of the same length.

There are $2m_0\, 2^{t-1}$ non-backtracking walks with t edges in N_0. Choose one of them uniformly. Then each of its steps is uniform over the $2m_0$ directed edges. So the expected length of the walk is $t(1 + k/m_0)$. At least half of the walks have length at most twice this value. Their number is at least

$$m_0\, 2^{t-1} \;\geq\; 12h(h-1) \;>\; n_0^2.$$

If one of these short walks visits a vertex twice, it contains a cycle that is not longer than the walk. Otherwise all of them are paths, and two of them have the same first vertex and the same last vertex. Their union contains a cycle of length at most the sum of their lengths. In both cases N has a cycle of length at most $4t(1 + k/m_0)$. □

Lemma 8.6 (Forest proportions at large girth)**.** *There are absolute constants $D_* \geq 3$ and C_0 with the following property. Let N be a pole of rank h with k ports and girth at least D. Suppose that*

(a) *$h \geq 256$, $k \leq 4D$ and $D \geq D_*$, or*

(b) *$h \geq \Lambda \geq 4$, $k \leq 2\Lambda$ and $D \geq C_0 \log(\Lambda + 2)$ for some number Λ.*

Then every set of ports of even size has forest proportion at most $\frac{3}{5}$.

Proof. If $k \leq 1$, the only set is the empty one, with forest proportion 2^{-h}. Let $k \geq 2$, and put $t = \lceil \log(8h) \rceil \leq \log h + 4$.

The rank is exponential in the girth. In case (a), $h - 1 > \frac{32}{3}t$: this holds at $h = 256$, where $t = 11$, and the ratio $(h-1)/(\log h + 4)$ increases with h. So Lemma 8.5 gives $D \leq 4t + 16tD/(3(h-1)) < 4t + D/2$. In case (b), $k/(3(h-1)) \leq 2h/(3(h-1)) < 1$, and Lemma 8.5 gives $D < 8t$. In both cases $D < 8t \leq 8 \log h + 32$, so

$$h \;\geq\; 2^{D/8-4}, \qquad \text{and } N \text{ has } n \geq h \text{ vertices.} \tag{8.2}$$

A ball without ports. Put $s = \lfloor D/16 \rfloor$. A ball of radius s has at most $3 \cdot 2^s$ vertices. So at most $3k\, 2^s$ vertices are within distance s of a port. In case (a) this is at most $12D\, 2^{D/16}$, and in case (b) at most $6\Lambda\, 2^{D/16}$. Choose D_* and C_0 so large that this number is less than $2^{D/8-4} \leq n$ in each case. Then some vertex w has distance more than s from every port. All vertices within distance s of w have degree three, and $2s < D$. So the ball of radius s around w is a tree in which every vertex at distance less than s from w has degree three, and it has at least $2^s \geq 2^{D/16-1}$ vertices. Fix a port

o. The distances from o to two vertices of this ball differ by at most $2s$. So the ball meets at most $2s+1 \le D$ layers with respect to o, and some layer has at least $2^{D/16-1}/D$ vertices.

Conclusion. Suppose that some set of ports has forest proportion more than $\frac{3}{5}$. By Lemma 8.4 every layer has at most $B(k+1)^2$ vertices. This is at most $B(4D+1)^2$ in case (a) and at most $B(2\Lambda+1)^2$ in case (b). In case (a) choose D_* so large that $2^{D/16-1}/D > B(4D+1)^2$ for $D \ge D_*$. In case (b) the function $2^{D/16}/D$ increases for $D \ge 32$, and at $D = C_0 \log(\Lambda+2)$ its ratio to $(\Lambda+2)^2$ is $(\Lambda+2)^{C_0/16-2}/(C_0 \log(\Lambda+2))$. This tends to infinity with C_0, uniformly in $\Lambda \ge 4$. So a large C_0 gives $2^{D/16-1}/D > B(2\Lambda+1)^2$. In both cases we have a contradiction. □

8.4 Two pieces

Lemma 8.7 (Product)**.** *Let G be a cubic multigraph, let P be a set of vertices, and let $U_1, U_2 \subseteq V(G) \setminus P$ be disjoint sets such that $G[U_1]$ and $G[U_2]$ are poles. Suppose that in each of these poles every set of ports of even size has forest proportion at most $\frac{3}{5}$. Then $q(G,P) \le \frac{9}{25}$.*

Proof. Let E' be the set of the edges of G that lie neither in $G[U_1]$ nor in $G[U_2]$. Fix a set $x \subseteq E'$ that is the restriction of some even subgraph. For $i = 1, 2$ let T_i be the set of the vertices of U_i that meet an odd number of edges of x. An edge set ξ with $\xi \cap E' = x$ is even if and only if $\xi \cap E(G[U_i])$ is a T_i-join of $G[U_i]$ for $i = 1, 2$: the degree condition at a vertex of U_i involves only edges of $G[U_i]$ and of x, and at the other vertices it involves only x. So, given $\xi \cap E' = x$, the two sets $\xi \cap E(G[U_i])$ are independent, and each is a uniform T_i-join. A vertex of degree three in $G[U_i]$ meets no edge of E', because G is cubic. So T_i is a set of ports, of even size.

If $\xi \cap E(G-P)$ is a forest, then $\xi \cap E(G[U_1])$ and $\xi \cap E(G[U_2])$ are forests. Given $\xi \cap E' = x$, this has probability at most $\frac{3}{5} \cdot \frac{3}{5}$. Average over x. □

Edges between U_1 and U_2 belong to E'. So the two sets may be adjacent.

Lemma 8.8 (The core of a piece)**.** *Let G be a cubic multigraph, and let X be a set of vertices such that $G[X]$ is connected and has rank $h \ge 2$. Then there is a set $U \subseteq X$ such that $G[U]$ is a pole of rank h with at most $|\delta X|$ ports, and the girth of $G[U]$ is at least the girth of $G[X]$.*

Proof. Delete vertices of degree one from $G[X]$ as long as there are any. The result $G[U]$ is connected, has the same cycles as $G[X]$ and all its degrees are two or three. When a vertex of degree one is deleted, two edges that left the set no longer do, and one edge inside the set now leaves it. So $|\delta U| \le |\delta X|$. A vertex of degree two in $G[U]$ has exactly one edge end outside $G[U]$, and a vertex of degree three has none. So $G[U]$ has $|\delta U|$ ports. □

Proposition 8.9 (Two pieces)**.** *Let G be a connected cubic multigraph and let P be a set of vertices. Let $X_1, X_2 \subseteq V(G) \setminus P$ be disjoint, nonempty and proper sets of vertices. Suppose that one of the following holds for $i = 1, 2$.*

(a) *$G[X_i]$ is connected of rank at least 256 and girth at least $D \ge D_*$, and $|\delta X_i| \le 4D$.*

(b) *$|X_i| \ge 8\Lambda^2$, $|\delta X_i| \le 2\Lambda$, and $G[X_i]$ has girth at least D, where $\Lambda \ge 4$ and $D \ge C_0 \log(\Lambda+2)$.*

Then $q(G,P) \le \frac{9}{25}$.

Proof. It suffices to find sets $U_i \subseteq X_i$ as in Lemma 8.7. In case (a), Lemma 8.8 gives a pole $G[U_i]$ of rank at least 256 with at most $4D$ ports and girth at least D, and Lemma 8.6(a) applies.

In case (b) write $n = |X_i|$, $d = |\delta X_i|$ and $c = \mathrm{comp}(G[X_i])$. Every component of $G[X_i]$ has an edge that leaves X_i, so $1 \le c \le d \le 2\Lambda$. Count the edge ends: $3n = 2|E(G[X_i])| + d$. So $\beta(G[X_i]) = (n-d)/2 + c$, and some component Y of $G[X_i]$ has rank at least

$$\frac{n-d}{2c} + 1 \;\ge\; \frac{8\Lambda^2 - 2\Lambda}{4\Lambda} + 1 \;>\; \Lambda.$$

Its boundary is a part of δX_i. Lemma 8.8 for Y gives a pole $G[U_i]$ of rank at least Λ with at most 2Λ ports and girth at least D, and Lemma 8.6(b) applies. □

9 Short cycles, without an excess

In the proof of Theorem 1.3, a maximal family of disjoint short cycles is small because of Lemma 6.4. That lemma bounds the family by a multiple of $1/\varepsilon$, where $\frac{1}{2} + \varepsilon$ is the assumed forest probability. For Theorem 1.5 we need a bound that holds as soon as the forest probability exceeds $\frac{1}{2}$. This section proves such a bound (Theorem 9.3). It uses Proposition 8.9(a): two clean pieces would give $\frac{9}{25}$. Throughout, $H = 256$.

9.1 Vertices that share a parity

Let Q be a connected multigraph without loops. For an edge e, the map $\xi \mapsto [e \in \xi]$ is a linear functional on $\mathcal{Z}(Q)$. For a vertex v let L_v be the space of functionals that is spanned by the edges at v. For a set A of vertices put $U_A = \sum_{a \in A} L_a$. We use one standard fact: if a set F of edges meets every even subgraph of Q in an even number of edges, then $F = \delta Y$ for some set Y of vertices. (Fix a spanning tree and a root. Let Y be the set of the vertices whose tree path from the root has an odd number of edges in F. Then F and δY contain the same tree edges, and the fundamental cycles show that they contain the same edges outside the tree.)

Lemma 9.1 (Strings)**.** *Let Q be a connected multigraph without loops, and let A be a nonempty proper set of its vertices. Put $b = |\delta A|$. Call a vertex $v \notin A$* relevant *if $L_v \cap U_A \neq 0$. There are a set of at most $2b$ relevant vertices, the* exceptions*, and at most $2b$ sequences of relevant vertices, the* strings*, with these properties. Every relevant vertex that is not an exception is in a string. If $v_1, \dots, v_m$ is a string and $1 \le i < j \le m$ with $j \ge i+2$, there is a set $M(i,j)$ of vertices such that*

(i) *$M(i,j)$ is disjoint from $A \cup \{v_i, v_j\}$, the graph $Q[M(i,j)]$ is connected, and $v_l \in M(i,j)$ for $i < l < j$;*

(ii) *every edge that leaves $M(i,j)$ ends in v_i or in v_j;*

(iii) *$M(i,j) \cap M(i',j') = \emptyset$ if $j \le i'$.*

Proof. Call the edges of δA *terminals*. The end of a terminal outside A is its *foot*. Every component of $Q - A$ contains a foot, because Q is connected.

Step 1. Let $v \notin A$ be relevant and not a foot, and let K be the component of $Q - A$ that contains v. We show that at least two components of $K - v$ contain feet. Let $0 \neq a \in L_v \cap U_A$. Then a is the sum of the functionals of a set F of edges at v, and also of a set F' of edges with an end in A. No edge is in both sets, because v is not a foot. The set $F \cup F'$ meets every even subgraph in an even number of edges, so $F \cup F' = \delta Y$ for some Y. We may assume that $v \notin Y$. No edge of $K - v$ is in $F \cup F'$. So each component of $K - v$ lies inside Y or outside Y, and F is the set of all edges from v to the components inside Y. For a component J of $K - v$, the cut

$\delta V(J)$ consists of the edges from J to v and the terminals with a foot in J. So on $\mathcal{Z}(Q)$ the sum of the edges from J to v equals the sum of the terminals with a foot in J. Hence a is the sum of the terminals with a foot in Y. The sum of all terminals with a foot in K is zero on $\mathcal{Z}(Q)$, because these terminals form the cut $\delta V(K)$. As $a \neq 0$, some component of $K - v$ inside Y contains a foot, and some component outside Y contains a foot.

Step 2. In each component K of $Q - A$ fix a spanning tree. For each terminal add a new leaf at its foot. Let T_K be the union of the paths of this tree between the new leaves, and let T be the union of the trees T_K. Every leaf of T is a new leaf, so T has at most b leaves and at most b vertices of degree at least three. The exceptions are the relevant vertices that are feet or have degree at least three in T. There are at most $2b$ of them.

Let v be a relevant vertex that is not an exception. By Step 1 two components of $K - v$ contain feet. The path of the tree between the two new leaves at these feet passes through v. So v is a vertex of T, and its degree in T is two. A *thread* is a maximal path of T whose inner vertices have degree two in T. There are at most $2b$ threads, and every vertex of degree two of T is an inner vertex of exactly one thread. The strings are the sequences of the relevant vertices that are not exceptions on one thread, in the order of the thread.

Step 3. Let v be a vertex of a string, in the component K. The forest $T - v$ has exactly two components that meet K, one on each side of v along its thread. Their vertex sets in K are the *near side* of v, towards the start of the thread, and the *far side* of v. Each side is connected in $K - v$, and all feet in K lie in the two sides. By Step 1 the two sides lie in different components of $K - v$.

Now let v_i and v_j be as in the lemma. Let Π be the part of their thread strictly between them. It contains v_{i+1}, it is connected, and it contains no foot, because a new leaf at a vertex of Π would give that vertex degree three in T. Let $M(i, j)$ be the vertex set of the component of $K - v_i - v_j$ that contains Π. This gives (i).

For (ii), suppose that $M(i, j)$ contains a foot w. Then w is a vertex of T outside Π. So w lies in the near side of v_i or in the far side of v_j; say in the near side of v_i. The path Π lies in the far side of v_i. But w and Π are joined in $K - v_i$ through $M(i, j)$, and the two sides of v_i lie in different components of $K - v_i$. This is a contradiction. So $M(i, j)$ contains no foot. An edge that leaves $M(i, j)$ ends in A, in v_i or in v_j, and it does not end in A.

For (iii), let $j \leq i'$. The set $M(i, j)$ is connected in $K - v_j$ and contains Π, which lies in the near side of v_j. So $M(i, j)$ lies in the component of $K - v_j$ that contains the near side of v_j. We show that $M(i', j')$ lies in the component of $K - v_j$ that contains the far side of v_j; then the two sets are disjoint. The set $M(i', j')$ is connected in $K - v_{i'}$ and contains the part Π' of the thread between $v_{i'}$ and $v_{j'}$, which lies in the far side of $v_{i'}$. If $j = i'$, this is the claim. If $j < i'$, then v_j lies in the near side of $v_{i'}$, so $v_j \notin M(i', j')$. Hence $M(i', j')$ is connected in $K - v_j$, and it contains Π', which lies in the far side of v_j. □

9.2 A packing bound

Proposition 9.2 (Packing)**.** *Let G be a connected cubic multigraph. Let P be a set of vertices that is empty or induces a connected subgraph, and let S be a set of s vertices outside P. Let $D \geq D_*$, and suppose that $G - (P \cup S)$ has girth at least D and that $q(G, P \cup S) > \frac{1}{2}$. Let $2 \leq d \leq 2D$, and let $X_1, \ldots, X_n$ be pairwise disjoint sets of vertices outside $P \cup S$ such that each $G[X_i]$ is connected and contains a cycle, and $|\delta X_i| \leq d$. Then*

$$n \;<\; (6H + 15)\, d\, (s + 1)\, 2^d.$$

Proof. Let $n \geq 1$.

The quotient. Contract each set X_i to a vertex x_i, and contract each component of $G - \bigcup_i X_i$ to a vertex. Delete the loops. Call the result Q, and call the vertices x_i *marked*. Then Q is connected, it has no loops, and x_i has degree $|\delta X_i| \le d$. The edges of Q are the edges of G between different contracted sets. As in the proof of Lemma 8.2(c), the map $\xi \mapsto \xi \cap E(Q)$ sends the uniform distribution on $\mathcal{Z}(G)$ to the uniform distribution on $\mathcal{Z}(Q)$. Given $\xi \cap E(Q)$, the sets $\xi \cap E(G[X_i])$ are independent, as in the proof of Lemma 8.7. Let I_i be the event that ξ contains no edge at x_i. On I_i the set $\xi \cap E(G[X_i])$ is a uniform even subgraph of $G[X_i]$. It is a forest only if it is empty, and this has probability at most $\frac{1}{2}$ because $G[X_i]$ contains a cycle. The sets X_i lie in $G - (P \cup S)$. Hence

$$\frac{1}{2} \;<\; q(G, P \cup S) \;\le\; \mathbb{E} \prod_{i=1}^{n} \Big(1 - \frac{\mathbf{1}_{I_i}}{2}\Big). \tag{9.1}$$

An independent family. Write $L_i = L_{x_i}$ for the space of Lemma 9.1 in Q. The edges at x_i form a cut, so their sum is zero on $\mathcal{Z}(Q)$ and $\dim L_i \le d - 1$. The event I_i says that the functionals of L_i vanish at ξ, so $\mathbb{P}(I_i) = 2^{-\dim L_i} \ge 2^{1-d}$. No L_i is zero, because then I_i would always hold, against (9.1). Let A be a maximal set of marked vertices whose spaces are linearly independent, and let $t = |A| \ge 1$. The events I_a, $a \in A$, are independent. By (9.1),

$$\frac{1}{2} \;<\; \prod_{a \in A} \Big(1 - \frac{\mathbb{P}(I_a)}{2}\Big) \;\le\; \big(1 - 2^{-d}\big)^t.$$

If $t \ge 2^d$, the right side is less than $\frac{1}{e}$. So $t < 2^d$. If all marked vertices are in A, we are done.

Strings. Otherwise A is a nonempty proper set of vertices of Q, with $b = |\delta A| \le dt$. By the maximality of A, every marked vertex outside A is relevant in the sense of Lemma 9.1. So all marked vertices outside A, except at most $2b$, lie on at most $2b$ strings.

Consider one string, and let $u_1, \dots, u_m$ be its marked vertices in the order of the string. Suppose that $m \ge (H+2)(s+3)$. For $0 \le l \le s+2$ let M_l be the set $M(\cdot,\cdot)$ of Lemma 9.1 for the two string vertices $u_{l(H+2)+1}$ and $u_{(l+1)(H+2)}$. By the lemma, the sets M_l are pairwise disjoint and connected in Q. Each contains H marked vertices, and every edge that leaves M_l ends in one of two marked vertices, so $|\delta M_l| \le 2d$. Let $\widehat{M}_l$ be the union of the sets of vertices of G that were contracted to the vertices of M_l. Then $G[\widehat{M}_l]$ is connected, and $|\delta \widehat{M}_l| \le 2d \le 4D$. It contains H of the pairwise disjoint sets X_i, each with a cycle, so its rank is at least H.

The sets $\widehat{M}_l$ are pairwise disjoint. If P is not empty, it is connected and disjoint from all X_i, so it lies in one contracted set. Hence at most one $\widehat{M}_l$ meets P. At most s of them meet S. So two of the $s+3$ sets avoid $P \cup S$. They induce subgraphs of $G - (P \cup S)$, which has girth at least $D \ge D_*$. By Proposition 8.9(a), for the set $P \cup S$, we get $q(G, P \cup S) \le \frac{9}{25}$. This contradicts (9.1). So $m < (H+2)(s+3)$.

Count. The marked vertices are in A, or exceptions, or on strings. Hence

$$n \;\le\; t + 2b + 2b\,(H+2)(s+3) \;\le\; t + (6H+14)(s+1)\,b \;\le\; (6H+15)\,d\,(s+1)\,t \;<\; (6H+15)\,d\,(s+1)\,2^d.$$

The second inequality is $2 + 2(H+2)(s+3) \le (6H+14)(s+1)$, which holds with equality for $s = 0$. □

9.3 Raising the girth

Theorem 9.3 (Raising the girth)**.** *There is an absolute constant K_1 with the following property. Let G be a connected cubic multigraph. Let P be a set of vertices that is empty or induces a connected subgraph. Suppose that $q(G, P) > \frac{1}{2}$ and that $G - P$ has girth at least D_*. Put $D_j = 2^j D_*$ for $j \ge 0$. Then there are sets $S_0 \subseteq S_1 \subseteq \cdots$ of vertices outside P such that, for every $j \ge 0$,*

(i) *S_j is the union of the vertex sets of a maximal family of pairwise disjoint cycles of length at most D_j in $G - P$; in particular $G - (P \cup S_j)$ has girth greater than D_j;*

(ii) $\log(|S_j| + 1) \le 2D_j + K_1 \log^2(D_j + 2)$.

Proof. Put $S_{-1} = \emptyset$. Let $j \ge 0$ and suppose that S_{j-1} is defined. Let $a = D_*$ if $j = 0$ and $a = D_{j-1} + 1$ if $j \ge 1$. Then $G - (P \cup S_{j-1})$ has girth at least $a \ge D_*$, and $b := D_j \le 2a$. Take a maximal family $\mathcal{Q}_j$ of pairwise disjoint cycles of length at most b in $G - (P \cup S_{j-1})$, and let S_j be the union of S_{j-1} and their vertex sets.

For a cycle C of the family, $G[V(C)]$ is connected and contains a cycle, and $|\delta V(C)| \le |C| \le b$, because G is cubic. Also $q(G, P \cup S_{j-1}) \ge q(G, P) > \frac{1}{2}$, because a forest in $G - P$ is a forest in a subgraph. So Proposition 9.2 applies with $S = S_{j-1}$, $D = a$ and $d = b$:

$$|\mathcal{Q}_j| \;<\; (6H + 15)\, b\, (|S_{j-1}| + 1)\, 2^b, \qquad |S_j| + 1 \;\le\; (6H + 16)\, b^2\, 2^b\, (|S_{j-1}| + 1).$$

(i) The families $\mathcal{Q}_0, \dots, \mathcal{Q}_j$ together are a family of pairwise disjoint cycles of length at most D_j in $G - P$. A further cycle of length at most D_j in $G - P$ that is disjoint from S_j would contradict the maximality of $\mathcal{Q}_j$.

(ii) Take logarithms in the recursion. Since $D_0 + \cdots + D_j < 2D_j$ and $j \le \log D_j$,

$$\log(|S_j| + 1) \;\le\; \sum_{i \le j} (D_i + 2 \log D_i + \log(6H + 16)) \;\le\; 2D_j + (\log D_j + 1)(2 \log D_j + 11). \qquad \square$$

The bound of Lemma 6.4 for the same family is about $D_j 2^{D_j}/\varepsilon$. Theorem 9.3 loses a factor 2^{D_j} and gains the independence of ε.

10 Links, a moment inequality and a chain

This section uses the notation of Section 6: regions, the events $\mathcal{O}_A$, and the number $\ell(A, B)$ of links. Its three lemmas on links and moments are formalised (Table 3).

10.1 Components between two connected sets

Lemma 6.9 bounds the number of cyclic components with a small boundary by Lemma 6.4, and that bound contains $1/\varepsilon$. For components that join the same two connected sets there is a bound without ε.

Lemma 10.1 (Parallel links)**.** *Let Γ be a connected multigraph, let A and B be disjoint regions, and let P be a set of vertices with $q(\Gamma, P) > \frac{1}{2}$. Let $t \ge 0$. Then at most 2^{t+1} components Y of $\Gamma - (A \cup B)$ have all of the following properties: Y has an edge to A and an edge to B; Y avoids P; Y contains a cycle; $|\delta Y| \le t$.*

Proof. Let $Y_0, Y_1, \dots, Y_n$ be such components. For $1 \le i < j \le n$ the graph $\Gamma - Y_i - Y_j$ is connected. Indeed, it contains the connected set $A \cup B \cup Y_0$, and every other component of $\Gamma - (A \cup B)$ is connected and has an edge to $A \cup B$. No edge joins Y_i and Y_j. So the regions Y_i and Y_j have exactly one link, and by Lemma 6.2

$$\mathbb{P}(\mathcal{O}_{Y_i} \cap \mathcal{O}_{Y_j}) = \mathbb{P}(\mathcal{O}_{Y_i})\, \mathbb{P}(\mathcal{O}_{Y_j}).$$

Let E_i be the event that $\mathcal{O}_{Y_i}$ holds and $\xi \cap E(\Gamma[Y_i])$ is not empty. By the last part of Lemma 6.2, the events E_i and E_j are independent as well, and $\mathbb{P}(E_i) \ge 2^{-t} \cdot \frac{1}{2}$. Let N be the number of the events

$E_1, \dots, E_n$ that occur, and $\nu = \mathbb{E}N \geq n\,2^{-t-1}$. Then $\mathbb{E}N^2 \leq \nu^2 + \nu$, and the Cauchy–Schwarz inequality gives

$$\mathbb{P}(N = 0) \;\leq\; 1 - \frac{\nu^2}{\mathbb{E}N^2} \;\leq\; \frac{1}{1+\nu}.$$

If $\xi \cap E(\Gamma - P)$ is a forest, then no E_i occurs, because Y_i avoids P and a nonempty even subgraph is not a forest. So $\frac{1}{2} < q(\Gamma, P) \leq 1/(1+\nu)$. Hence $\nu < 1$ and $n < 2^{t+1}$. □

The components between A and B are in parallel. One parity condition couples all of them, and it does not couple any family that omits one. The value $\frac{1}{2}$ in Lemma 6.4 comes from regions in series.

Lemma 10.2 (Links, without an excess)**.** *Let G be a connected cubic multigraph, and let P be a nonempty set of vertices such that $G[P]$ is connected and $q(G, P) > \frac{1}{2}$. Let S be the union of the vertex sets of a maximal family of pairwise disjoint cycles of length at most D in $G - P$, and put $J = G - (P \cup S)$. Let $8 \leq D \leq L$ and $t \geq 1$ with $2(t + 2\lfloor D/8 \rfloor) \leq D$. Then any two disjoint cycles C, C' of J of length at most L satisfy*

$$\ell(V(C), V(C')) \;\leq\; 1 + \frac{2L}{t+1} + 2^{t+1} + 578\Big(\frac{L}{D}\Big)^2.$$

Proof. This is the proof of Lemma 6.9 with one change. Put $U = V(C) \cup V(C')$. At most $2L$ edges leave U. At most one component of $G - U$ meets P. At most $2L/(t+1)$ components have more than t edges to U. Of the other components that are joined to both cycles, at most 2^{t+1} contain a cycle, by Lemma 10.1 with $A = V(C)$ and $B = V(C')$. This is the change. The remaining components that are joined to both cycles are trees with at most t edges to U. As in the proof of Lemma 6.9, each of them lies in J and gives a path of J from C to C' with at most t edges; an edge between the cycles is such a path as well. Divide each cycle into at most $17L/D$ arcs of at most $\lfloor D/8 \rfloor$ vertices. Two of the paths with ends in the same pair of arcs would give a cycle of J of length at most $2t + 4\lfloor D/8 \rfloor \leq D$. So there are at most $289(L/D)^2$ such paths. □

10.2 Pairs with exactly two links

Let G, P, S, J, D and L be as in Lemma 10.2, with $L > D$, and put $s = \lfloor (D-1)/2 \rfloor$. Let $\mathcal{C}$ be the set of the cycles of J of length at most L, and define a_C, X_C, Z and μ as in Section 6, in the cubic multigraph G in the place of Γ. The graph J is simple, has maximum degree three and girth greater than D, so Lemma 6.6 holds for it. It gives

$$\sum_{C \in \mathcal{C},\; C \ni w} a_C \;\leq\; L\,2^{-s} \qquad \text{for every vertex } w. \tag{10.1}$$

Lemma 10.3 (Two links)**.** *Suppose that $\ell(V(C), V(C')) \leq b$ for all disjoint $C, C' \in \mathcal{C}$, and put $\eta = 2^b L^3 2^{-s}$. Let W_2 be the sum of $a_C a_{C'}$ over the ordered pairs of disjoint cycles of $\mathcal{C}$ with exactly two links. If $\mu > 0$, then*

$$W_2 \;>\; \mu^2 - (1+\eta)\,\mu.$$

Proof. By (6.4), a disjoint pair with ℓ links has $\mathbb{E}(X_C X_{C'}) = a_C a_{C'} 2^{\ell-1}$. This is at most $a_C a_{C'}$ for one link, and it is $2a_C a_{C'}$ for two links. Pairs with at least three links are treated as in the proof of Theorem 1.3: for a fixed C they contribute at most $a_C\, 2^b L^3 2^{-s} = \eta\, a_C$. Intersecting pairs contribute nothing, and $\mathbb{E}X_C^2 \leq a_C$. Hence

$$\mathbb{E}Z^2 \;\leq\; \mu^2 + W_2 + (1+\eta)\,\mu.$$

If $\xi \cap E(G-P)$ is a forest, then $Z = 0$. So $\mathbb{P}(Z \neq 0) < \frac{1}{2}$, and the Cauchy–Schwarz inequality gives $\mu^2 \leq \mathbb{E}Z^2\,\mathbb{P}(Z \neq 0) < \frac{1}{2}\,\mathbb{E}Z^2$. □

In the proof of Theorem 1.3 the same computation ends with an upper bound for μ in terms of $1/\varepsilon$. Here it is read in the other direction, and it needs only $q(G,P) > \frac{1}{2}$: almost all pairs, by weight, have exactly two links.

10.3 Separators are nested

Lemma 10.4 (Nested separators)**.** *Let u and p be two vertices of a connected graph H. Let C_1 and C_2 be disjoint connected sets of vertices that avoid u and p, such that u and p lie in different components of $H - C_i$ for $i = 1, 2$. Let U_i be the vertex set of the component of u in $H - C_i$. Then exactly one of the following holds: $C_1 \subseteq U_2$ and $U_1 \subseteq U_2$; or $C_2 \subseteq U_1$ and $U_2 \subseteq U_1$.*

Proof. Take a path from u to p, and let z be its last vertex before the first vertex in C_1. The part of the path up to z lies in U_1. In the same way there is a path from p to a vertex next to C_1 that avoids C_1. The set C_2 is connected and disjoint from C_1, so it lies in one component of $H - C_1$. If this component is neither U_1 nor the component of p, then the two partial paths and a path inside C_1 form a walk from u to p that avoids C_2. This is impossible. So C_2 lies in U_1 or in the component of p in $H - C_1$, and the same holds with the indices exchanged.

Suppose that C_2 lies in the component of p in $H - C_1$. Then C_2 avoids U_1. So every vertex of U_1 is joined to u outside C_2, and so is every vertex of C_1, through z and the connected set C_1. Hence $U_1 \cup C_1 \subseteq U_2$. If C_2 lies in U_1 and C_1 lies in U_2, then the path from p to a vertex next to C_1, a path inside C_1, and a path inside U_2 from C_1 to u give a walk from p to u outside C_2. This is impossible. So if C_2 lies in U_1, then C_1 lies in the component of p in $H - C_2$, and $U_2 \cup C_2 \subseteq U_1$ by the first case with the indices exchanged. The two alternatives exclude each other, because $C_1 \subseteq U_2$ and $U_2 \subseteq U_1$ would give $C_1 \subseteq U_1$. □

So a family of pairwise disjoint connected sets that separate u from p is linearly ordered: write $C_1 < C_2$ in the first case of the lemma. The order is transitive, because it is the order of the sets U_i by inclusion.

Lemma 10.5 (Blocks)**.** *In the situation of Lemma 10.4, let $C_1 < C_2$ and put $X = U_2 \setminus (U_1 \cup C_1)$. Then every edge of H that leaves X ends in C_1 or in C_2. If $C_1 < C < C_2$ for a third set C of the family, then $C \subseteq X$. If $C_1 < C_2 \leq C_3 < C_4$, then the sets $U_2 \setminus (U_1 \cup C_1)$ and $U_4 \setminus (U_3 \cup C_3)$ are disjoint.*

Proof. An edge that leaves U_2 ends in C_2, because U_2 is a component of $H - C_2$. No edge joins X to U_1: its end in X would lie in the component U_1 of $H - C_1$. So an edge from X to $U_1 \cup C_1$ ends in C_1. For the second statement, $C \subseteq U_2$, and C avoids $U_1 \cup C_1$ because $U_1 \cup C_1 \subseteq U_C$, the component of u in $H - C$. For the third, $U_2 \subseteq U_3$. □

10.4 The chain

Lemma 10.6 (Chain)**.** *In the situation of Lemma 10.3, suppose that*

$$\mu > 20, \qquad \eta \leq \frac{1}{2}, \qquad L^2 2^{-s} \leq \frac{1}{2}.$$

Fix a vertex $p \in P$. Then there are a cycle $C \in \mathcal{C}$, a vertex $u \notin V(C)$ that has a neighbour on C, and pairwise disjoint cycles $C_1, \dots, C_T \in \mathcal{C}$ with

$$T \;\geq\; \frac{8 \cdot 2^s}{L^3},$$

such that each C_i is disjoint from C and avoids u, and u and p lie in the same component of $G - V(C)$ and in different components of $G - V(C) - V(C_i)$.

Proof. For disjoint $C, C' \in \mathcal{C}$ say that C' is *far behind* C if C' does not lie in the component of p in $G - V(C)$. This relation is antisymmetric. Let C' be far behind C. A path from p to its first vertex on C stays in the component of p in $G - V(C)$ before it ends. So it avoids C'. As C is connected and disjoint from C', all of C lies in the component of p in $G - V(C')$. So C is not far behind C'. Hence the ordered pairs (C, C') with C' far behind C have total weight $\sum a_C a_{C'} \le \mu^2/2$.

Fix C. Every vertex of C has exactly one edge end outside C, so at most L vertices outside C have a neighbour on C. A cycle C' that is joined to C by an edge passes through one of them. By (10.1) these cycles have total weight at most $L^2 2^{-s} \le \frac{1}{2}$. So the ordered pairs of disjoint cycles that are joined by an edge have total weight at most $\mu/2$.

Call an ordered pair (C, C') *good* if the cycles are disjoint, have exactly two links, are not joined by an edge, and C' is not far behind C. By Lemma 10.3 and the two bounds above, the good pairs have total weight more than

$$\mu^2 - \tfrac{3}{2}\mu - \tfrac{1}{2}\mu^2 - \tfrac{1}{2}\mu \;=\; \mu(\tfrac{1}{2}\mu - 2) \;>\; 8\mu.$$

So for some $C \in \mathcal{C}$ the cycles C' with (C, C') good have total weight more than 8. Fix this C, and put $H = G - V(C)$.

Let (C, C') be good. The two links are components Y_1, Y_2 of $G - V(C) - V(C')$ that are joined to both cycles, because no edge joins the cycles. The vertex p lies in at most one of them. So one of them, say Y, avoids p. Let u be a vertex of Y with a neighbour on C. The cycle C' lies in the component of p in H, and Y is connected and joined to C'. So u lies in that component as well. The set Y is a component of $H - V(C')$ that contains u and not p. So C' separates u from p in H.

There are at most L choices for u. So for some vertex u the cycles C' that separate u from p in H, as above, have total weight at least $8/L$. Take a maximal family $C_1, \dots, C_T$ of pairwise disjoint cycles among them. Every cycle of the collection meets one of the at most TL vertices of the family. By (10.1),

$$\frac{8}{L} \;\le\; TL \cdot L\, 2^{-s}. \qquad \square$$

11 Proof of Theorem 1.5

Let D_* and C_0 be as in Lemma 8.6, and let K_1 be as in Theorem 9.3. Let G and P be as in Theorem 1.5, and suppose that $q(G, P) > \frac{1}{2}$. We derive a contradiction when r is large. All $o(\cdot)$ refer to $r \to \infty$ and are uniform in G and P. Put

$$x = \log r, \qquad z = \log x, \qquad A = 6\, e^{168}, \qquad L = \lfloor Ax \rfloor, \qquad t = \lfloor z - 2 \log z \rfloor.$$

Two girth scales. Let $D_j = 2^j D_*$ and let $S_0' \subseteq S_1' \subseteq \cdots$ be the sets of Theorem 9.3. Let D^- be the first D_j that is at least $\sqrt{x}$, and let D^+ be the first D_j that is at least $x/16$. For large r,

$$\sqrt{x} \le D^- < 2\sqrt{x}, \qquad \frac{x}{16} \le D^+ < \frac{x}{8}, \qquad D^- < D^+.$$

Let $S^- \subseteq S^+$ be the corresponding sets, and put $J^\pm = G - (P \cup S^\pm)$. Then J^- has girth greater than D^-, J^+ has girth greater than D^+, and $J^+ \subseteq J^-$. By Theorem 9.3(ii),

$$|S^-| \;\le\; 2^{4\sqrt{x} + K_1 \log^2(2\sqrt{x}+2)} = r^{o(1)}, \qquad |S^+| \;\le\; 2^{x/4 + K_1 \log^2(x)} = r^{1/4 + o(1)}. \tag{11.1}$$

The small scale D^- will make the final pieces clean. The large scale D^+ gives a long chain.

Cycles at the large scale. We apply Section 10 with $S = S^+$, $J = J^+$, $D = D^+$ and $s = \lfloor (D^+ - 1)/2 \rfloor \geq x/32 - 2$. The hypotheses of Lemma 10.2 hold for large r: $8 \leq D^+ \leq L$, and $2(t + 2\lfloor D^+/8 \rfloor) \leq D^+$ because $t \leq z \leq D^+/4$. So every two disjoint cycles of J^+ of length at most L have at most

$$b \;=\; 1 + \frac{2L}{t+1} + 2^{t+1} + 578\Big(\frac{L}{D^+}\Big)^2 \;\leq\; 1 + \frac{2Ax}{z - 2\log z} + \frac{2x}{z^2} + 578\,(16A)^2 \;\leq\; \frac{3Ax}{z}$$

links. Hence

$$\eta = 2^b L^3 2^{-s} \;\leq\; r^{3A/z}\,(Ax)^3\,4\,r^{-1/32} = r^{-1/32+o(1)} = o(1), \qquad L^2 2^{-s} = o(1).$$

The graph J^+ is simple, with maximum degree three and girth greater than D^+. It has $v = 2r - 2 - |P| - |S^+| = 2r - o(r)$ vertices, and its deficit is $d_J \leq 3(|P| + |S^+|) = r^{1/4+o(1)}$. Let μ be the total weight of its cycles of length at most L. The proof of Lemma 6.8 applies to J^+ with these parameters. It gives $\mu \geq (1 - o(1))\,T$, and for even ℓ with $6x \leq \ell \leq L$

$$2^{-\ell}\,\mathrm{tr}\,B^\ell \;\geq\; 1 - \frac{L\,d_J}{2v} - 3v\,r^{-3} = 1 - o(1).$$

The sum of $1/(2\ell)$ over the even ℓ in $[6x, L]$ is $(\frac{1}{4} - o(1))\ln(A/6) = 42 - o(1)$. So $\mu > 20$ for large r.

The chain. Fix $p \in P$. By Lemma 10.6 there are a cycle C of J^+ of length at most L, a vertex u, and pairwise disjoint cycles $C_1, \ldots, C_T$ of J^+ of length at most L with

$$T \;\geq\; \frac{8 \cdot 2^s}{L^3} \;\geq\; \frac{2}{A^3} \cdot \frac{r^{1/32}}{x^3},$$

such that each C_i separates u from p in $G - V(C)$. Let H be the component of $G - V(C)$ that contains u and p. Each $V(C_i)$ is connected and separates u from p, so it lies in H. By Lemma 10.4 we may number the cycles so that $C_1 < C_2 < \cdots < C_T$. Let U_i be the vertex set of the component of u in $H - V(C_i)$.

Blocks. Put $w = \lceil 128A^2x \rceil + 2$. For $l = 0, 1, \ldots$ with $(l+1)w \leq T - 1$ let

$$X_l \;=\; U_{(l+1)w+1} \setminus (U_{lw+1} \cup V(C_{lw+1})).$$

There are at least $\lfloor (T-1)/w \rfloor \geq r^{1/32}/x^5$ such sets, for large r. By Lemma 10.5 they are pairwise disjoint. Each contains the $w - 1$ cycles between its two end cycles. These cycles are pairwise disjoint and each has more than D^+ vertices, so

$$|X_l| \;\geq\; (w-1)(D^+ + 1) \;\geq\; 128A^2x \cdot \frac{x}{16} = 8A^2x^2 \;\geq\; 8L^2.$$

Every edge of H that leaves X_l ends on one of its two end cycles. An end cycle has at most L vertices, and each of them has one edge end outside the cycle. So at most $2L$ edges of H leave X_l.

Clean blocks. Discard a set X_l if it contains a vertex with a neighbour on C, or a vertex of S^-. At most L vertices outside C have a neighbour on C, so this discards at most $L + |S^-| = r^{o(1)}$ sets. For large r at least two sets remain. Let X be one of them.

- X avoids P. The set P is connected and disjoint from the cycles, and it contains p, which is not in U_i for any i. A path in $G[P]$ from p to a vertex of U_i would have to pass through C_i or through C.

- No edge of G joins X to $V(C)$, and H is a component of $G - V(C)$. So the edges of G that leave X are edges of H, and $|\delta X| \le 2L$.
- X avoids $P \cup S^-$, so $G[X]$ is a subgraph of J^-, and its girth is greater than D^-.

The contradiction. The two remaining sets satisfy the hypotheses of Proposition 8.9(b) with $\Lambda = L$ and $D = D^-$: they are disjoint, they avoid P, each has at least $8L^2$ vertices and at most $2L$ boundary edges, and $D^- \ge \sqrt{x} \ge C_0 \log(L+2)$ for large r. So $q(G,P) \le \frac{9}{25}$. This contradicts $q(G,P) > \frac{1}{2}$. Hence there is R such that $q(G,P) \le \frac{1}{2}$ whenever $r \ge R$. □

Remark 11.1. The set X is defined in a component H of $G - V(C)$, and not in J^+. So it contains the vertices of $S^+ \setminus S^-$ that lie between its two end cycles. This is the reason for the two scales. The chain needs the girth of J^+, and the pieces need only the girth of J^-, which has far fewer deleted vertices than the chain has cycles.

Remark 11.2. The same chain proves a quantitative form of Theorem 1.3 for graphs with short cycles. Suppose that $q(\Gamma, P) > \frac{1}{2} + \varepsilon$ in the setting of Section 6. Delete short cycles at two scales with Lemma 6.4, find the chain, and count the weight of the cycles inside the clean blocks. This weight is at most $(1 + o(1))/(4\varepsilon)$ by the second moment, and it is at least a constant times the number of clean blocks. The result is $q(\Gamma, P) \le \frac{1}{2} + r^{-\kappa}$ for every fixed $\kappa < \frac{1}{3}$, when $|P| \le r/\log^2 r$ and $\Gamma[P]$ has at most $\gamma \log r$ components, for a constant $\gamma = \gamma(\kappa) > 0$. Both models checked this proof; it is in the repository as a research note [22] and it is not printed here. With an exponential form of Lemma 6.4, Astra's report [22] improves the error term to $O((\log r)^{11/2}/\sqrt{r})$. Fable 5.2 has not checked that last refinement line by line.

12 The two auxiliary theorems

Theorems 7.1 and 7.2 are due to Astra. Their proofs are in the supplement [19, 20, 21], which has about sixty pages. This section states what they depend on and describes the arguments. It proves nothing.

Status. The supplement consists of research reports that Astra wrote during the work, and that it checked with several independent passes of its own. Fable 5.2 read the report [20], and it checked the proof of Theorem 12.1 in [21]. Of [19] it read the statements. It did not check the case analysis of [19] line by line. None of the supplement is formalised. The reports contain exact tests of their finite identities on small graphs. A reader who wants to verify Theorem 1.1 must read the supplement.

No circle. The supplement uses Theorem 1.3 and Theorem 1.2. It uses neither Theorem 1.5 nor Theorem 1.1.

12.1 Paths between two vertices

For two vertices s, t of a connected multigraph N let $P_{st}(N)$ be the number of paths from s to t. By Theorem 1.6(a), $P_{st}(N) \le 2^{\beta(N)}$. Cycles through an edge are paths: if $e = st$ is an edge of G, then $c_e(G) = P_{st}(G - e)$. So Lemma 2.3 reads $c(G) = c(G - e) + P_{st}(G - e)$. The supplement studies graphs in which P_{st} is more than half of its largest possible value.

Theorem 12.1 (Stability of the number of paths; supplement [21]). *For every $\varepsilon \in (0, \frac{1}{2}]$ there is a constant C_ε with the following property. Let N be a connected graph of rank b and let $s \neq t$ be two of its vertices. If $P_{st}(N) \geq (\frac{1}{2} + \varepsilon)2^b$, then every path from s to t has at least $b - C_\varepsilon$ edges.*

This is a stable form of Theorem 1.6(a), which is the case of equality with $C = 0$. Its proof uses Theorem 1.3 with a fixed error, through the localisation lemma and a version of Lemma 8.3 for the layers around a shortest path. The first version, in which every path has at least $c(\varepsilon)\, b$ edges, is due to Fable 5.2. The additive form is due to Astra. A second result of [19] is a gap: $P_{st}(N) = 2^b$ or $P_{st}(N) \leq \frac{15}{16}2^b$.

12.2 Small cyclic cuts

Let G be a cubic multigraph with a cyclic cut δX of size $k \leq 4$. Cut the k edges. Each side becomes a multigraph with k vertices of degree two, the *ports*. A cycle of G lies on one side, or it crosses the cut. A crossing cycle meets each side in a set of disjoint paths between ports. So the number of crossing cycles is a sum, over the ways to pair the ports, of products of two numbers: the number of such systems of paths on one side and on the other side. The proof of Theorem 7.1 bounds these numbers.

For $k = 2$ there is one pair of ports on each side. The crossing cycles number $P_{st}(N_1)P_{st}(N_2)$. If both factors are at most about half of their maximum, the product is at most about $\frac{1}{4}\, 2^{r-1}$, and Theorem 1.2 bounds the cycles inside the two sides. If one factor is large, Theorem 12.1 applies to that side. It shows that the side is thin: its vertices lie in layers of bounded size around a shortest path. A count along the layers then shows that such a side has only $O(2^{b/2})$ cycles. This is the *path–cycle tradeoff* of [19].

For $k = 3$ and $k = 4$ there are several pairs of ports, and for $k = 4$ also systems of two paths. The report [19] proves inequalities between these counts on one side, classifies the sides for which the counts are near their maxima, and evaluates the sum for each type. In the extremal type both sides are chains of four-cycles, as in the graphs of Horak and Kim. There the largest terms of the sum cancel exactly, and the remaining term has the order $r2^{r/2}$. The count of pairs of disjoint cycles in a thin side is the source of the factor r.

12.3 Bounded girth

Let G be a cubic multigraph without loops, of girth $g \geq 5$, with no cyclic cut of size at most four. Let $C = v_0v_1 \cdots v_{g-1}$ be a shortest cycle and $e = v_0v_{g-1}$. By Lemma 2.3, $c(G) = c(G-e)+P_{v_0v_{g-1}}(G-e)$, and $c(G-e) \leq (\frac{1}{2} + o(1))2^{r-1}$ by Theorem 1.2. The report [20] sorts the paths from v_0 to v_{g-1} in $G - e$ by the vertex v_i at which they leave the path $v_0v_1 \cdots v_{g-1}$. After the deletion of $v_0, \ldots, v_i$, the rest of the path lies in a multigraph of rank at most $r - 2 - i$. If it had more than $(\frac{1}{2} + \varepsilon)$ of the largest possible number of paths, a consequence of Theorem 12.1 would give a cyclic cut of size at most four in G. So the branch at v_i contributes at most about $\frac{1}{2}\, 2^{r-2-i}$ paths. The sum over $i < g - 1$ is a geometric series that falls short of $\frac{1}{2}\, 2^{r-1}$ by 2^{r-g-1}. This gives

$$c(G) \;\leq\; 2^{r-1} - 2^{r-g-1} + o_g(2^r),$$

which is Theorem 7.2 for each fixed girth. For girth at most four, a shortest cycle itself gives a cyclic cut of size at most four when $r \geq 9$.

The deficit 2^{r-g-1} decreases with the girth. So this argument cannot treat graphs whose girth grows with r. Theorem 1.5 treats them.

13 The graphs of Horak and Kim

The lower bound (1.1) is due to Horak and Kim [10, Theorem 5.1]. We recall their graphs for odd rank, because they show where the second term comes from.

Let $j \geq 2$. Take vertices a_i, b_i, c_i, d_i for i modulo j. Join each of a_i, b_i to each of c_i, d_i; these four edges are the *square* i. Join c_i to a_{i+1} and d_i to b_{i+1}; these two edges are the *interface* i. The graph M_j is cubic, with $4j$ vertices, $6j$ edges and rank $r = 2j + 1$. In the notation of [10] it is the graph MC_{2j}.

Proposition 13.1 (Horak and Kim)**.** *For $j \geq 2$,*

$$c(M_j) \;=\; 2^{2j} + (2j + \tfrac{1}{2})2^j - 3j \;=\; 2^{r-1} + (r - \tfrac{1}{2})2^{(r-1)/2} - \tfrac{3}{2}(r-1).$$

Proof. The two edges of an interface form a cut together with the two edges of any other interface. So a cycle uses an odd number of edges of every interface, or an even number of edges of every interface.

One edge of every interface. There are 2^j ways to choose the edges. In each square the cycle then runs from one of a_i, b_i to one of c_i, d_i, and there are two such paths: the edge, and the path through the other two vertices of the square. Every choice gives a cycle. This gives $2^j \cdot 2^j = 2^{r-1}$ cycles.

An even number of edges of every interface. If the cycle uses no interface, it is one of the j squares. If it uses both edges of all interfaces, it uses two disjoint edges of every square, and there are two ways to choose them in each square. Such a choice is one cycle, and not two, for half of the 2^j choices: 2^{j-1} cycles. Otherwise the interfaces that it uses are d consecutive ones, with $1 \leq d \leq j-1$. There are j such sets for each d. The cycle closes in the two squares at the ends, in two ways each, and it uses two disjoint edges in each of the $d-1$ squares between them. This gives $j\,2^{d+1}$ cycles for each d. Together,

$$j + 2^{j-1} + j\sum_{d=1}^{j-1} 2^{d+1} \;=\; (2j + \tfrac{1}{2})2^j - 3j. \qquad \square$$

The first class contains exactly half of the even subgraphs of M_j, and all of them are cycles. The second term counts the cycles in the other half. Most of these are the cycles that use a proper set of consecutive interfaces. They exist because each square is a cycle with a cut of four edges around it. This is the configuration that Theorem 7.1 treats.

The graphs M_j also show that the conclusion of Theorem 1.5 fails without the hypothesis on the girth. Let W be a square of M_j and let e be one of its edges. The cycles of the first class through e number 2^{r-2}, and the square itself is a further cycle through e. So $c_e(M_j) > 2^{r-2}$, and Lemma 2.3 gives $q(M_j, V(W)) > \frac{1}{2}$. Here $M_j - V(W)$ contains cycles of length four.

For even rank, Horak and Kim use the same construction on two cycles of odd length, with one uncrossed pair of rungs [10, Section 5]. We tested both formulas of (1.1) by computer for small ranks (Appendix B).

14 Two elementary theorems

This section proves Theorem 1.6. Part (a) is Proposition 14.1, and part (b) is Proposition 14.2. Both are formalised (Table 1).

Three conditions on a subgraph W of G are easily confused, so we separate them first. Let W have $m \geq 1$ edges.

(i) *Every outside part is a forest.* This means $p(G, W) = 1$. It holds if and only if $G - E(W)$ is a forest: a subset of a forest is a forest, and a cycle of $G - E(W)$ is its own outside part.

(ii) *No outside part is wasted.* This means that every outside part is the outside part of some cycle. It is the hypothesis of Proposition 14.2, and it does not imply (i). Let G be K_5 minus an edge, and let W be the triangle on its three vertices of degree four. Then $G - E(W)$ is $K_{2,3}$. The sixteen outside parts are the empty set, twelve paths and three four-cycles. Each of them is the outside part of a cycle, but only thirteen are forests. So no outside part is wasted, and $p(G, W) = 13/16$.

(iii) *Equality in* (1.3). This never holds. The cycles with empty outside part lie in W, so they have fewer than m different lengths. Every other outside part accounts for at most m lengths. So the cycles of G have fewer than $m\,2^{r-t}$ lengths.

Proposition 14.1 is about a fourth condition: a graph has the largest possible number of paths between two vertices.

14.1 Two terminals

Proposition 14.1. *Let N be a connected graph and let $u \neq v$ be two of its vertices. Then N has at most $2^{\beta(N)}$ paths from u to v. If it has exactly $2^{\beta(N)}$, then $N - E(A)$ is a forest for every path A from u to v, and every such path has at least $\beta(N)$ edges.*

Proof. A path from u to v is a $\{u, v\}$-join in the sense of Example 2.2: exactly u and v have odd degree in it. The joins form a coset of $\mathcal{Z}(N)$, so there are $2^{\beta(N)}$ of them. This proves the first statement. Now suppose that all joins are paths, and let A be one of them. If $N - E(A)$ contained a cycle C, then $A \cup C = A + C$ would be a join that is not a path. So $N - E(A)$ is a forest. Deleting an edge lowers the rank by at most one, so $\beta(N) \leq |A|$. □

Equivalently, an edge f of a graph G lies on at most $2^{\beta(G)-1}$ cycles, and if it lies on that many, then each of them has at least $\beta(G)$ edges. Equality in Proposition 14.1 holds for the necklace N_j of Section 5 between v_0 and v_j. It also holds for K_4 minus an edge and for $K_{3,3}$ minus an edge, in each case between the ends of the deleted edge.

This explains the shape of the construction. The necklace N_k turns k units of rank into 2^k path lengths, the most that k units can give. By Proposition 14.1, no gadget that does this has a path with fewer than k edges between its terminals. So the cycles through N_k have length greater than k, and the lengths $3, \ldots, k$ must be produced elsewhere. The construction uses a necklace with about $\log k$ beads for them. Its own short lengths require a third necklace, and so on.

Each of these necklaces costs a factor of two. Consider a necklace N_j that is closed by a chord f_j, together with f_j and the chords of the smaller necklaces. Call this part W. The rest of the graph is attached to W at the two vertices v_0 and v_j only. By Example 2.2, half of the outside parts are even subgraphs of the rest. A cycle that uses an edge of W cannot have a nonempty even subgraph as its outside part. So nearly half of the outside parts are of no use to the cycles through W. These $\log^* k$ halvings are the term $\log^* n$ of Theorem 1.4.

Formal counterpart. Proposition 14.1 is the theorem `twoTerminalTheorem`. The formal statement also covers the case $u = v$. The formal proof gets the bound $\beta(N) \leq |A|$ from a count: an even subgraph of N is determined by its edges on the path A.

14.2 No waste

Lemma 2.1 counts the cycles of G by their outside parts. Recall that an outside part is *wasted* if it is not the outside part of any cycle. A graph cannot avoid all waste for long.

Proposition 14.2. *Let G be a connected graph and let $W \subseteq G$ be a subgraph that contains a cycle. Suppose that no outside part is wasted. Then every nonempty even subgraph of $G - E(W)$ is a cycle, and*

$$\beta(G) - \beta(W) \;\leq\; |V(W)| + 3.$$

Proof. Let z be a nonempty even subgraph of $G - E(W)$. Then z is its own outside part. By assumption some cycle C has outside part z, so $z \subseteq C$. A cycle that contains a nonempty even subgraph is equal to it. So z is a cycle.

In particular, two edge-disjoint cycles of $G - E(W)$ would have a union that is an even subgraph but not a cycle. So the graph $G' = G - E(W)$, on the vertex set $V(G)$, has no two edge-disjoint cycles. By Lemma 14.3 below, $|E(G')| \leq |V(G)| + 3$. Now $\beta(G) = |E(G)| - |V(G)| + 1$ and $\beta(W) \geq |E(W)| - |V(W)| + 1$. Hence $\beta(G) - \beta(W) \leq |E(G')| - |V(G)| + |V(W)| \leq |V(W)| + 3$. □

Lemma 14.3 (Erdős and Pósa [7])**.** *A multigraph with n vertices and at least $n + 4$ edges contains two edge-disjoint cycles.*

Proof. We include the short proof. Let H be a counterexample with $|V(H)| + |E(H)|$ minimal, and let n be its number of vertices. Then H has exactly $n + 4$ edges and no vertex of degree at most one. It has no vertex of degree two either. If such a vertex carries a loop, then the rest of H contains a second cycle. Otherwise the vertex and its two edges can be replaced by a single edge, which gives a smaller counterexample. So $2(n + 4) \geq 3n$, that is, $n \leq 8$. Let C be a shortest cycle of H. Since $H - E(C)$ is a forest, $n + 4 - |C| \leq n - 1$, so $|C| \geq 5$. Thus H is a simple graph of minimum degree at least 3 and girth at least 5, and such a graph has at least $1 + 3 + 6 = 10$ vertices, a contradiction. □

A graph in which every nonempty even subgraph is a cycle has $2^\beta - 1$ cycles, the largest possible number. Such graphs are known. Entringer and Slater [6, Theorem 1] quote a theorem of Maurer, and of Mateti and Deo: a connected graph has this property if and only if it reduces to K_1, K_3, K_4 minus an edge, K_4 or $K_{3,3}$. A reduction deletes a vertex of degree one, or suppresses a vertex of degree two whose two neighbours are not adjacent. So the first conclusion of Proposition 14.2 puts $G - E(W)$ into a known class. In the language of codes, a binary code in which every nonzero word is minimal is intersecting [1, §2].

The second conclusion is sharp up to the additive constant. If W is a cycle $w_1 \cdots w_m$ and G consists of W and a path of length two from w_i to w_{i+1} for every i modulo m, then no outside part is wasted and $\beta(G) - \beta(W) = |V(W)|$. In the example of (ii) above, $\beta(G) - \beta(W) = |V(W)| + 1$. An exhaustive search found nothing better for a triangle W in graphs with at most eight vertices.

Formal counterpart. Proposition 14.2 is the theorem `noWasteTheorem`. There W is given by its edge set S, and $\beta(W)$ is defined through the graph on all vertices of G with edge set S. Lemma 14.3 is the theorem `ExactCounting.Multi.card_le_of_intersecting`. It is stated for multigraphs in the language of even edge sets: if any two nonempty even edge sets share an edge, then there are at most $n + 3$ edges. The formal proof uses the same reductions as the proof above. In the last step it finds both even sets by a pigeonhole count.

15 Problems

An earlier draft of this paper asked three questions. All three now have answers.

Is $\psi(r) \le 2^{r-1} + 2^{(1-c)r}$ for some $c > 0$, and is the error term in (1.2) at most r^{-c}? Yes. Theorem 1.1 gives $c = \frac{1}{2}$ up to the factor r, which is best possible by (1.1). Remark 11.2 gives the error term $r^{-\kappa}$ for $\kappa < \frac{1}{3}$.

If a connected graph N has at least $(\frac{1}{2} + \varepsilon)2^{\beta(N)}$ paths between two vertices, must each of these paths have at least $c(\varepsilon)\beta(N)$ edges? Yes, and even $\beta(N) - C_\varepsilon$ edges (Theorem 12.1).

Can the constant 3 in Proposition 14.2 be replaced by 1? Yes, and 1 is best possible. Astra proved this, and Fable 5.2 checked the proof; it is in the research notes [23] and is not printed here.

The constant. By Theorem 1.1 and (1.1), the numbers

$$\kappa(r) = \frac{\psi(r) - 2^{r-1}}{r\, 2^{r/2}}$$

are bounded, with $\liminf \kappa(r) \ge \frac{1}{\sqrt{2}}$ along odd r and $\liminf \kappa(r) \ge \frac{1}{2}$ along even r. The exact values $\psi(r)$ for $r \le 12$ [16] give $\kappa(12) > 1$, so small ranks do not show the limit. The constant K of Theorem 1.1 is not explicit.

Problem 15.1. Does $\kappa(r)$ converge along the odd r and along the even r? Are the limits $\frac{1}{\sqrt{2}}$ and $\frac{1}{2}$? For large r, are the graphs of Horak and Kim the graphs with the most cycles?

While this paper was written, Astra wrote a report [24] that claims a positive answer: for all large r, equality holds in (1.1). The report builds on Theorem 1.1 and on a finer analysis of the cuts of size four. Fable 5.2 has not checked it, and this paper does not use it.

Problem 15.2. Find an explicit pair (K, r_0) such that $\psi(r) \le 2^{r-1} + Kr2^{r/2}$ for all $r \ge r_0$. Find any explicit bound below $\frac{15}{16}2^r$ at ranks that can be written down.

When is the proportion at most one half? Theorem 1.5 needs large girth outside P. The graphs M_j show that some condition is necessary. The proof of Theorem 1.1 suggests that small cyclic cuts are the only obstruction.

Problem 15.3. Let G be a cubic graph of large rank r with no cyclic cut of size at most four, and let W be a shortest cycle. Is $q(G, V(W)) \le \frac{1}{2}$? Is every edge of W on at most 2^{r-2} cycles?

An edge on at most half of the cycles. Entringer and Slater [6] conjectured that a graph has an edge that lies on at most half of its cycles; the graphs K_4 and $K_{3,3}$ are exceptions. This compares $c_e(G)$ with $c(G)$, and our results compare it with 2^{r-1}. So they do not decide the conjecture. In a census of the connected cubic graphs with at most 22 vertices the conjecture holds, with the exceptions K_4 and $K_{3,3}$, and with little room: the least value of $c(G) - 2\min_e c_e(G)$ is 14 at 22 vertices. They also conjectured that *every vertex* of a graph of minimum degree at least three, other than K_4 and $K_{3,3}$, has an edge that lies on at most half of the cycles. This is false. Take K_4 and replace three of its vertices by triangles. The result has 10 vertices and 47 cycles, and each edge at the fourth vertex lies on 24 of them. We do not know whether this example is new; Guichard [9] studied these conjectures, and we have not seen that paper.

Problem 15.4. Does every connected cubic graph other than K_4 and $K_{3,3}$ have an edge that lies on at most half of its cycles?

Pancyclic graphs. Theorem 1.4 leaves the constant A. The lower bound in its proof starts at a rank R_0 that is not explicit.

Problem 15.5. Determine lim sup and lim inf of $h(n) - \log n - \log^* n$.

A The formal statements

The Lean 4 code is in the repository [13]. This paper refers to commit `a5cb5fa` on the branch `feat/cycle-count-bound`. The code uses Lean v4.19.0 and a pinned revision of Mathlib. It has 274 modules and 46,306 lines. The nine modules in `Erdos1016/CycleCount/` contain the proof of Theorem 1.2. The eight modules in `Erdos1016/ExactCounting/` contain the proof of Theorem 1.6. The 10 modules in `Erdos1016/PolynomialBound/`, with 2,215 lines, contain the formalised steps of the proof of Theorem 1.5 (Table 3). The rest is the formalisation from [11].

A.1 The statements behind Theorem 1.2

The following text is copied from `Erdos1016/CycleCount/Statement.lean`, `Asymptotic.lean` and `Erdos1016/Main.lean`, without the comments and the proofs.

```
def cycleEdgeSets {V : Type*} (G : SimpleGraph V) : Set (Set (Sym2 V)) :=
  {s | ∃ (v : V) (p : G.Walk v v), p.IsCycle ∧ s = {a | a ∈ p.edges}}

def numCycles {V : Type*} (G : SimpleGraph V) : ℕ := (cycleEdgeSets G).ncard

def CycleCountStatement : Prop :=
  ∃ r₁ : ℕ, ∀ (n : ℕ) (G : SimpleGraph (Fin n)) (r : ℕ),
    r + n = Nat.card G.edgeSet + Nat.card G.ConnectedComponent → r₁ ≤ r →
    (numCycles G : ℝ) ≤
      (1 / 2 + 21 / Real.logb 2 (Real.logb 2 (Real.logb 2 (r : ℝ)))) * 2 ^ r

def cycleCounts (r : ℕ) : Set ℕ :=
  {c | ∃ (n : ℕ) (G : SimpleGraph (Fin n)),
    r + n = Nat.card G.edgeSet + Nat.card G.ConnectedComponent ∧ numCycles G = c}

def maxCycles (r : ℕ) : ℕ := sSup (cycleCounts r)

def AsymptoticCycleCount : Prop :=
  Tendsto (fun r : ℕ => (maxCycles r : ℝ) / 2 ^ r) atTop (𝓝 (1 / 2))

theorem cycleCountTheorem : CycleCount.CycleCountStatement
theorem cycleCountAsymptotic : CycleCount.AsymptoticCycleCount
```

Four remarks help to compare this with Theorem 1.2.

- `SimpleGraph`, `Walk` and `IsCycle` are Mathlib's. A walk with `IsCycle` is a closed walk of length at least 3 that repeats no vertex except its start. The set `cycleEdgeSets G` contains the edge set of each such walk once. So `numCycles G` is the number of cycles in the sense of this paper, and not the number of closed walks.
- The hypothesis `r + n = ...` says that $r = \beta(G)$. It avoids subtraction of natural numbers.
- The threshold is existentially quantified, as r_1 is in the text. In the proof it is $\max(r_0, 2^{2048})$, where r_0 is the threshold of the forest estimate.

- In Lean, `sSup` of an empty or unbounded set of natural numbers is 0. This does not make `maxCycles` degenerate. The code proves that `cycleCounts r` is bounded by 2^r (`cycleCounts_bddAbove`) and that its supremum is attained for $r \geq 3$ (`maxCycles_mem`).

The theorem `cycleCountTheorem_general` states the same bound for simple graphs on every finite vertex type.

A.2 The statements behind Theorem 1.6

The following text is copied from `Erdos1016/ExactCounting/TwoTerminals.lean`, `NoWaste.lean` and `Erdos1016/Main.lean`, without the comments and the proofs.

```
def pathEdgeSets {V : Type*} (G : SimpleGraph V) (u v : V) : Set (Set (Sym2 V)) :=
  {s | ∃ p : G.Walk u v, p.IsPath ∧ s = {a | a ∈ p.edges}}

def numPaths {V : Type*} (G : SimpleGraph V) (u v : V) : ℕ := (pathEdgeSets G u v).ncard

def TwoTerminalStatement : Prop :=
  ∀ (n : ℕ) (G : SimpleGraph (Fin n)) (r : ℕ) (u v : Fin n),
    G.Connected → r + n = Nat.card G.edgeSet + 1 →
      numPaths G u v ≤ 2 ^ r ∧
      (numPaths G u v = 2 ^ r →
        ∀ p : G.Walk u v, p.IsPath →
          r ≤ p.length ∧ (G.deleteEdges {a | a ∈ p.edges}).IsAcyclic)

def IsEvenEdgeSet {V : Type*} (s : Set (Sym2 V)) : Prop :=
  ∀ v : V, Even {e | e ∈ s ∧ v ∈ e}.ncard

def NoWasteStatement : Prop :=
  ∀ (n : ℕ) (G : SimpleGraph (Fin n)) (S : Set (Sym2 (Fin n))) (r t : ℕ),
    G.Connected → S ⊆ G.edgeSet →
    r + n = Nat.card G.edgeSet + 1 →
    t + n = S.ncard + Nat.card (SimpleGraph.fromEdgeSet S).ConnectedComponent →
    (∀ s : Set (Sym2 (Fin n)), s ⊆ G.edgeSet → IsEvenEdgeSet s →
      ∃ c ∈ CycleCount.cycleEdgeSets G, c \ S = s \ S) →
    (∀ c₁ ∈ CycleCount.cycleEdgeSets (G.deleteEdges S),
      ∀ c₂ ∈ CycleCount.cycleEdgeSets (G.deleteEdges S), ¬ Disjoint c₁ c₂) ∧
    r ≤ t + {v | ∃ e ∈ S, v ∈ e}.ncard + 3

theorem twoTerminalTheorem : ExactCounting.TwoTerminalStatement
theorem noWasteTheorem : ExactCounting.NoWasteStatement
```

Five remarks help to compare this with Theorem 1.6.

- A walk with `IsPath` repeats no vertex. A path is counted once, by its edge set.
- The graph `G.deleteEdges s` is G without the edges in `s`, and `IsAcyclic` says that it is a forest.
- The first statement does not assume $u \neq v$.
- In the second statement `S` is $E(W)$. The line with `t` says that t is the rank of the graph with edge set S on all n vertices. Isolated vertices do not change the rank, so $t = \beta(W)$. The set in the last line is $V(W)$.

- Theorem 1.6(b) assumes that W contains a cycle. The formal statement does not need this hypothesis: it follows from the main hypothesis for the empty even set.

A.3 The forest estimate

Theorem 1.3 is the theorem `ShortProof.fewComponentForestEstimate`, which has the following type.

```
def FewComponentForestEstimate : Prop :=
  ∃ rmin : ℕ, ∀ (G : PhysicalGraph), G.IsConnected → rmin ≤ G.cycleRank →
    ∀ F : Finset G.CycleWord, F.Nonempty →
      (witnessSupportEdges G F).card ^ 2 ≤ G.cycleRank →
      100 * (Nat.card (cycleUnionGraph G F).ConnectedComponent : ℝ) ≤
        Real.logb 2 (G.cycleRank : ℝ) →
      outsideForestProbability G (witnessSupportEdges G F) ≤
        (1 / 2 : ℝ) + 20 / tripleLog G.cycleRank
```

Here `PhysicalGraph` is the project's type of finite simple graphs with numbered vertices and edges. The family `F` is a nonempty finite set of cycles of `G`, and W is their union. The set `witnessSupportEdges G F` is $E(W)$. The graph `cycleUnionGraph G F` is W on the vertices that the cycles meet. The number `outsideForestProbability G E` is the number of even subgraphs x such that $x \setminus E$ is a forest, divided by $2^{\beta(G)}$. Finally `tripleLog r` is $\log^{(3)} r$. The statement of Theorem 1.4 in Lean is described in [11, §1.2].

A.4 Repeating the check

The repository contains a verifier. It rebuilds every module from source. It compares the types of the endpoint theorems with statements that are written out a second time in `audits/TheoremAudit.lean`. It prints the axioms of the audited declarations, among them all endpoints in Table 1 and the declarations in Table 3, and it accepts only `propext`, `Classical.choice` and `Quot.sound`. It rejects proofs that contain `sorry`. The commands are:

```
git clone https://github.com/JWKNT/erdos1016.git && cd erdos1016
git checkout a5cb5fa
elan toolchain install leanprover/lean4:v4.19.0
lake exe cache get
python3 -m unittest discover -s scripts -p 'test_*.py'
python3 scripts/verify.py --fresh --jobs 2
```

On 11 November 2026 the check passed on a machine of the authors for the Lean sources, the audit and the scripts of this commit, without the option `-fresh`. The continuous integration of the repository repeated the check on the same day, with a fresh build, for commit `fcea625`. That commit differs from `a5cb5fa` only in the folder `paper/cycle-count` and in four text files with research notes, so the Lean sources, the audit and the scripts are the same. GitHub Actions run 42980562548 records the result.

A.5 Steps of the proofs

Table 2 names the Lean declaration for each step of the formalised theorems. Namespaces are omitted, and files are relative to `Erdos1016/`. The rows for Section 3 match the text closely, with four differences. The formal proof of Lemma 3.1 uses no spanning tree: it chooses the edges of D

one at a time, and it records only that each of them lies in an even subgraph. The formal proof uses $r \le \beta(G^+) \le r + a$ in place of the equality. It states Lemma 3.2 together with the count of the cycles that avoid D. It writes k as $\lfloor \log \lfloor \log r \rfloor \rfloor + 1$, which is the same number.

The rows for Section 14 match the text, with the differences that are stated at the end of Sections 14.1 and 14.2.

The rows for Sections 4 to 6 are taken from [11, Appendix A]. We checked that these declarations exist at the commit above. We did not compare each of them with our wording. In these parts the formal proof replaces O-terms by explicit constants, and intermediate estimates differ in detail from the text. The statements of Theorems 1.3 and 1.4, of Proposition 4.2 and of Corollary 4.3 agree with the Lean statements.

Table 2: Steps of the formalised proofs and the corresponding Lean declarations.

In this paper	Declaration	File
Lemma 3.1	`exists_independent_edges`	`CycleCount/IndependentEdges`
Second neighbours	`exists_apex`	`CycleCount/IndependentEdges`
Chords and triangles	`addChords`, `cycleRank_addChords_le`, `exists_triangle_family`	`CycleCount/ChordExtension`, `CycleCount/TriangleFamily`
Lemma 3.2	`card_cycleWord_le`	`CycleCount/ConnectedBound`
Theorem 1.2, parameters	`witness_parameters`, `cycleCountBound_of_forestEstimate`	`CycleCount/Bound`
Theorem 1.2, Mathlib form	`cycleCountStatement_of_bound`	`CycleCount/Statement`
Lower bound $\psi(r) \geq 2^{r-1}$	`exists_graph_many_cycles`	`CycleCount/LowerBound`
$\psi(r) \sim 2^{r-1}$	`asymptoticCycleCount_of_statement`	`CycleCount/Asymptotic`
Lemma 4.1	`exists_budgeted_witness`	`Extremal/Witness/CapacityBudget`
Proposition 4.2	`recurrence_of_few_component_forest_estimate`	`Extremal/Recurrence/WitnessReduction`
Corollary 4.3	`recurrence_logStar_bound`	`Extremal/Recurrence/LinearExponentialDecay`
Proposition 5.1	`upper_bound_integer`	`Extremal/UpperBound`
Lemma 6.2	`multigraph_zeroCut_pair`	`CycleSpace/Graphical/MultigraphCutCorrelation`
Lemma 6.3	`exceptional_partner_card_le`	`CycleSpace/Graphical/DisjointRegionCuts`
Lemma 6.4	`smallCut_forest_bound`	`Probability/Moments/SmallCutRegionBound`
Lemma 6.5	`exists_marked_subcubic_reduction`	`Graph/Cubicization/MarkedReduction`
Lemma 6.7	`normalized_even_trace_lower`	`Nonbacktracking/Spectrum/DeficitTrace`
Lemma 6.8	`cycleWordMass_ge_log_ratio`	`Cycles/Counting/LowDegreeTraceCycles`
Lemma 6.9	`actual_cycle_links_le`	`Cycles/Geometry/ActualCycleLinkBound`
Theorem 1.3	`second_moment_le`, `forest_le_half_add_twenty`, `fewComponentForestEstimate`	`Probability/Moments/SeedFamilyAvoidance`, `Probability/Avoidance/FewComponentForest`
Paths and cycles in Section 14	`par_edges_of_isTrail`, `eq_empty_of_subset_path`, `eq_edges_of_subset_cycle`	`ExactCounting/EdgeParity`
Proposition 14.1	`card_even_le_two_pow`, `twoTerminalStatement`	`ExactCounting/EvenSets`, `ExactCounting/TwoTerminals`
Lemma 14.3	`exists_even_subset`, `intersecting_suppress`, `exists_small_dense`, `card_le_of_intersecting`	`ExactCounting/Multigraph`, `Suppression`, `ShortCycle`, `DisjointCycles`
Proposition 14.2	`noWasteStatement`	`ExactCounting/NoWaste`

A.6 The formalised steps of Theorem 1.5

Theorems 1.5 and 1.1 are not formalised. Table 3 lists the steps of the proof of Theorem 1.5 that are. Each row is a complete Lean proof of the statement named. The files are in `Erdos1016/PolynomialBound/`.

Table 3: The formalised steps of the proof of Theorem 1.5.

In this paper	Declaration	File
Lemma 8.7	`forest_density_le_mul`	`PolynomialBound/Product`
Lemma 10.1	`none_density_le`, `links_card_le_one`, `smallCyclicPart_card_le`	`PolynomialBound/ParallelLinks`
Lemma 10.2	`cycle_links_le`	`PolynomialBound/LinkBound`
Lemma 10.3	`second_moment_le`, `twoLink_weight_ge`	`PolynomialBound/TwoLinkMoment`
Lemma 10.4	`nested`, `before_total`, `before_asymm`, `before_trans`	`PolynomialBound/Separation`
Lemma 10.5	`mem_separator_of_adj_block`, `subset_block`, `block_not_adj`, `exists_chain`, `exists_blocks`	`PolynomialBound/Separation`, `PolynomialBound/Chain`
Lemma 10.6, the two weight bounds	`antisymm_weight_le`, `exists_good_row`	`PolynomialBound/Averaging`

The formal statements differ from the text in these points. Lemma 8.7 is formalised for two arbitrary disjoint vertex sets of a multigraph: if, for every even subgraph x and for $i = 1, 2$, the sets $x + z$ with $z \in \mathcal{Z}(G[U_i])$ are forests inside $G[U_i]$ in a proportion of at most a_i, then the forest probability is at most $a_1 a_2$. Lemma 10.2 is formalised for the graph and the marked set of Section 6; its first term is the number of components of the marked set, which is one in Lemma 10.2. Lemma 10.3 is formalised with $\geq$ in the place of $>$, and for a forest probability of at least $\frac{1}{2}$. The separators of Lemma 10.4 are formalised for arbitrary connected sets in a simple graph. The step of Lemma 10.6 from two links to a separator, the pieces of Section 8 other than Lemma 8.7, and Section 9 are not formalised.

Three further modules in the same folder prove a conditional statement. If the forest estimate holds with the error term $r^{-\kappa}$ for marked sets with few components (`PolynomialMarkedBound`), then every graph of large rank r has at most $(\frac{1}{2}+r^{-\kappa'})2^r$ cycles (`polynomialCycleCountStatement_of_markedBound`). The hypothesis is the statement of Remark 11.2. It is not formalised.

B Tests on small graphs

We tested the examples of this paper, and our wording of the proofs, on small graphs. The scripts are in the folder `paper/cycle-count/paper-checks` of the repository [13]. They use exact arithmetic, except for the two scripts on the non-backtracking matrix, which compute eigenvalues in floating point. All tests passed. A passed test is evidence, not a proof.

- *The examples* (`check_examples.py`). The numbers in Example 2.2, the ring of four-cycles in Section 3 for $2 \leq j \leq 7$, the example in (ii) of Section 14, and the cycle with ears in Section 14 for $3 \leq m \leq 6$.
- *Section 3* (`check_chords.py`). 300 random connected graphs with 4 to 8 vertices and every admissible k, which gives 857 cases. The script takes D outside a random spanning tree and

builds the chords, G^+ and W. It checks Lemmas 3.1 and 3.2, the existence of the second neighbours, and the stated properties of G^+ and W.

- *Lemma 2.1 and Proposition 14.2* (`check_counting.py`). 4000 random connected graphs with 4 to 8 vertices, each with a random subgraph W. The script checks the number of outside parts, the bound of Lemma 2.1(b), and the bound $2^r p + 2^{r-t}(1-p)$ on the number of all cycles, which follows from the proof of that lemma. In the 2027 cases without a wasted outside part it checks both conclusions of Proposition 14.2.

- *Proposition 14.2, exhaustively* (`nowaste_exh.py`, `nowaste_exh2.py`). The scripts enumerate connected graphs G that contain a fixed triangle or a fixed four-cycle W, with up to 8 vertices. Without a wasted outside part, the largest value of $\beta(G) - \beta(W) - |V(W)|$ was 1 for the triangle and 0 for the four-cycle.

- *Proposition 5.1* (`check_construction.py`). The graph of Section 5 for all $n \le 139$ and for $n \in \{263, 264, 265, 300, 500, 520, 521, 522\}$. Each graph is pancyclic, and the number of chords never exceeds $\lfloor \log n \rfloor + \log^* n + 1$.

- *Lemmas 6.2 and 6.3* (`check_links.py`). 400 random connected multigraphs with 5 to 9 vertices and 5221 pairs of disjoint regions with at most three vertices each. The script checks the link formula and the bound on exceptional partners.

- *Lemmas 6.6 and 6.7* (`check_nb.py`, `check_nb2.py`). 292 random graphs of maximum degree 3 with 10 to 40 vertices, and the Petersen, Heawood and Tutte–Coxeter graphs. The scripts check the walk bound, the two statements on eigenvalues, and the trace inequality for even $\ell \le 24$.

- *Lemma 2.3 and Section 13* (`check_cyclic_squares.py`). The graphs M_j for $2 \le j \le 7$ and the graphs of Horak and Kim of even rank from 4 to 14. The script counts the cycles by a search and, for the smaller graphs, by a list of all even subgraphs. It compares the counts with (1.1) and with Proposition 13.1. For a square W of M_j it computes $q(M_j, V(W))$ and the number of cycles through an edge of W, and it checks Lemma 2.3 and the inequality $q(M_j, V(W)) > \frac{1}{2}$.

- *Lemma 10.1* (`check_parallel_links.py`). 300 random connected multigraphs with two connected sets A, B and two to four components that join them. The script lists all even subgraphs. It checks that the events $\mathcal{O}_Y$ are mutually independent for every family of components that omits one link (2228 families), and that two components that are the only links have the factor two.

- *Lemmas 8.2, 8.7 and 9.1* (`check_pieces.py`). Small random connected multigraphs. The script checks the rank formula and the equality of distributions in Lemma 8.2 in 400 cases. It checks the product structure in the proof of Lemma 8.7 for 6973 sets x in 300 graphs. For Lemma 9.1 it computes the relevant vertices by linear algebra, follows the construction of the proof in 4000 cases, and checks the two bounds and the three properties of the sets $M(i, j)$; the cases contain 1132 strings with at least three vertices and 7103 sets $M(i, j)$.

The reports of the supplement describe further tests of their finite identities [18, 19, 20]. The census of the cubic graphs with at most 22 vertices that Section 15 quotes is in the research notes [23]. Its values of $\psi(r)$ agree with [16].

C Sources

We compared the bibliographic data of every reference with the publisher's record, with Crossref or with arXiv, except where we say otherwise. For [6] we used the scan that is linked from the OEIS. We read [1], [6], [8], [11], [16], and the parts of [10] that concern cycles. Of [2] we read the abstract. We have not seen [3], [5], [7], [9] or [14]. What we say about [5] is taken from [4] and [11]. What we say about [9] is taken from [16], and we did not compare its bibliographic data with a record. No proof in this paper depends on a source that we have not seen. The lemma from [7] is proved in Section 14, and the facts related to [3, 14] are proved in Section 6.

We looked for an earlier proof of $\psi(r) \sim 2^{r-1}$, for any bound below $\frac{15}{16}2^r$ for cubic graphs, and for an upper bound of the form $2^{r-1} + o(2^r)$ with an explicit error term. We used web searches and the literature discussed in [1], [10] and [16], and we found none. A search of this kind can miss things.

The two theorems of Section 14 are elementary, and they are close to known results. Lemma 14.3 is due to Erdős and Pósa. The graphs in which every nonempty even subgraph is a cycle were characterised by Maurer and by Mateti and Deo; we know this from [6, Theorem 1] and have not seen the original papers. We do not know whether Proposition 14.1 or the bound in Proposition 14.2 has been stated before.

The supplement. The entries [18] to [23] are files in the folder `paper/cycle-count/research/2026-11-10` of the repository [13]. They are research reports and notes, written by the two models during the work. They are not publications, and no person has reviewed them. The reports [18] to [22] have compiled versions with the same name and the ending `.pdf`. The same folder contains the other notes of this work, among them results that this paper does not use. The repository also keeps the earlier drafts of this paper, next to the manuscript in the folder `paper/cycle-count`.

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[23] Astra and Fable 5.2, research notes, November 2026. Files `no-waste-phase2.tex` (the constant in Proposition 14.2) and `fable-census.md` (cubic graphs with at most 22 vertices) in the supplement.

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